

Methods for Analytical Validation

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Learning objectives

- Describe the process of analytical validation
- Explain basic concepts and metrics
- Develop a study for evaluating and comparing algorithms
- Assess the results of an evaluation

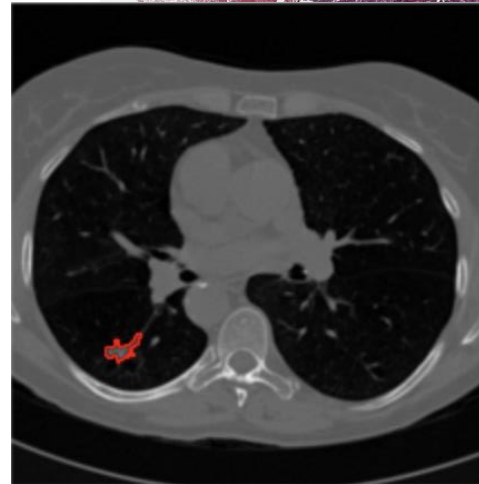
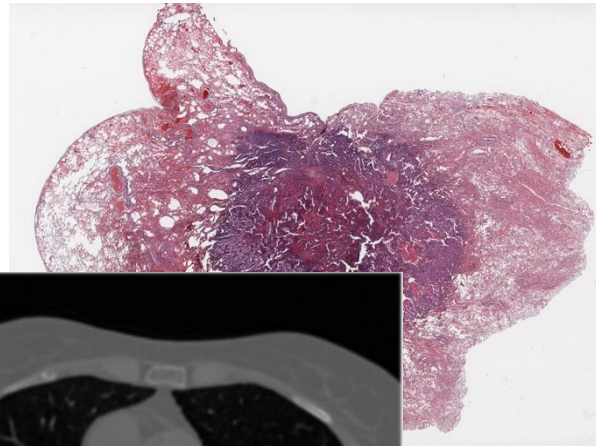
Course notes and supplemental materials are available at

<https://uclawillhsu.github.io/spie-tutorial>

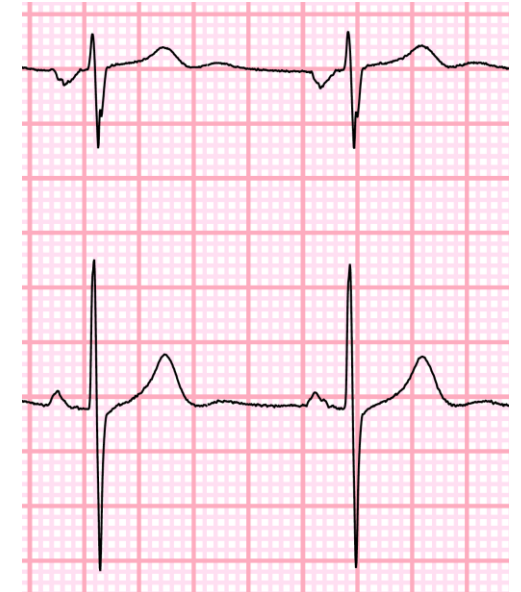
Learning from varied data types...

Patient ID	Chest Pain	Myocardial Infarct
1234	YES	YES
1235	NO	NO
1236	YES	NO
1237	YES	YES
1238	NO	NO

Textual / numerical

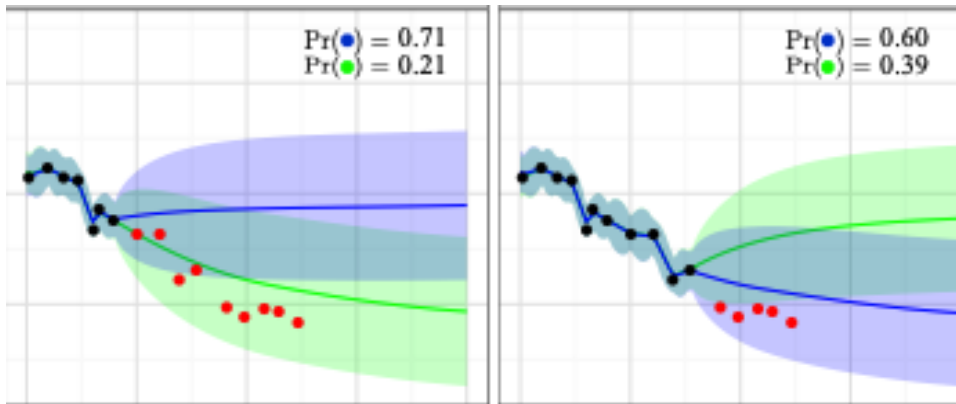
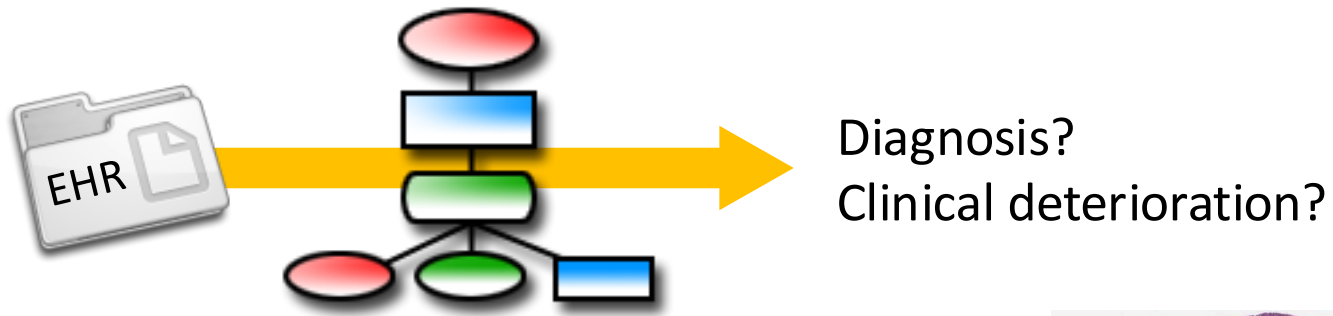


Imaging

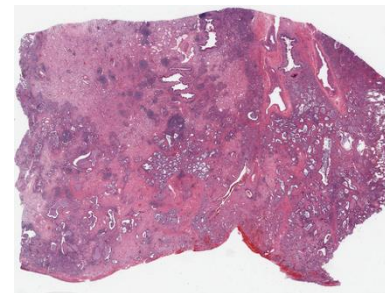


Waveform

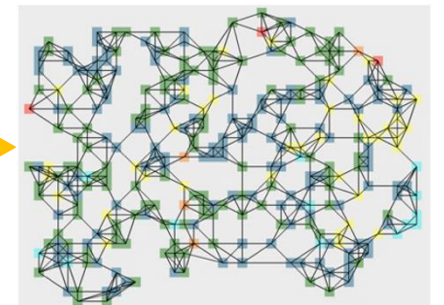
... to provide actionable insights

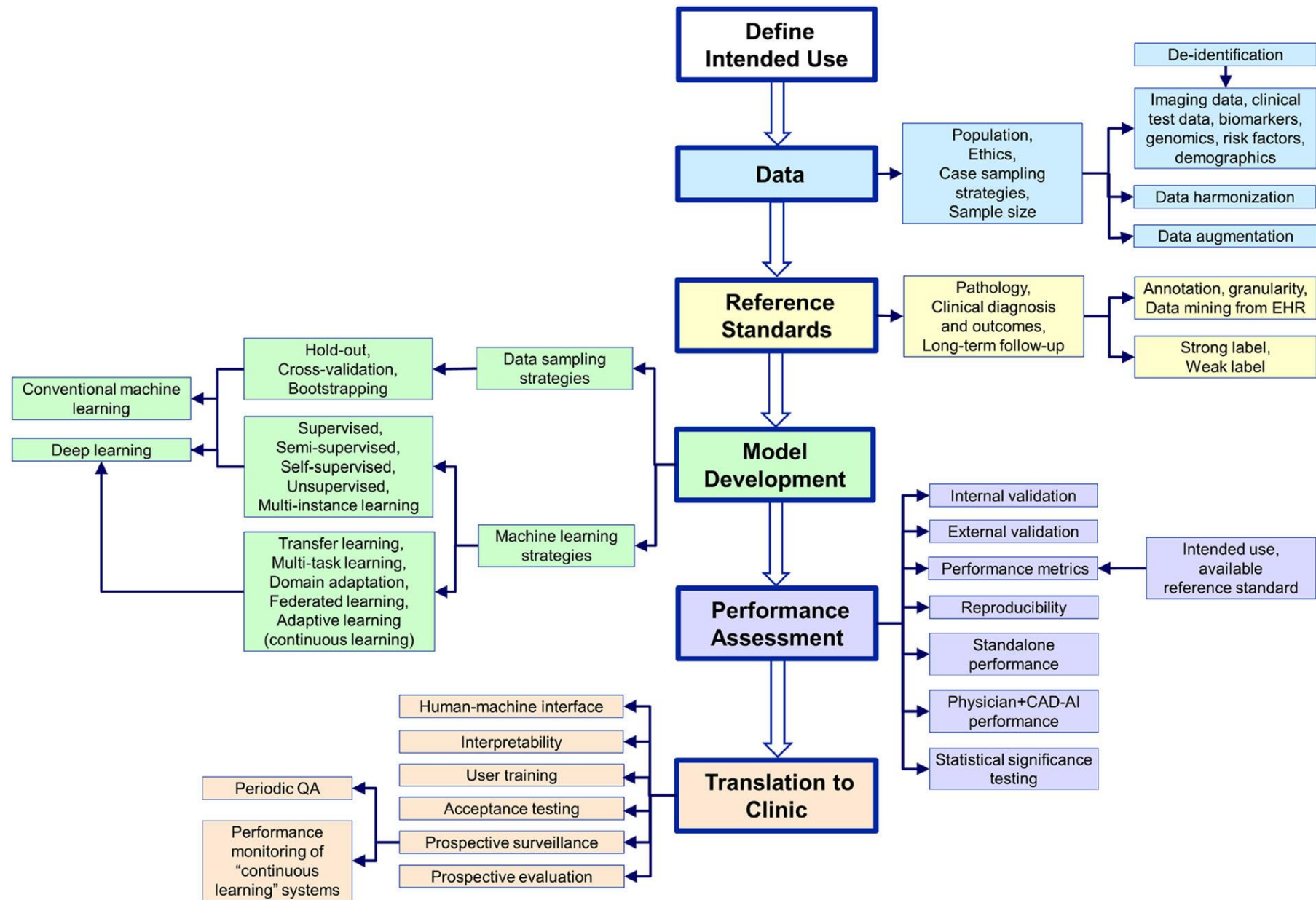


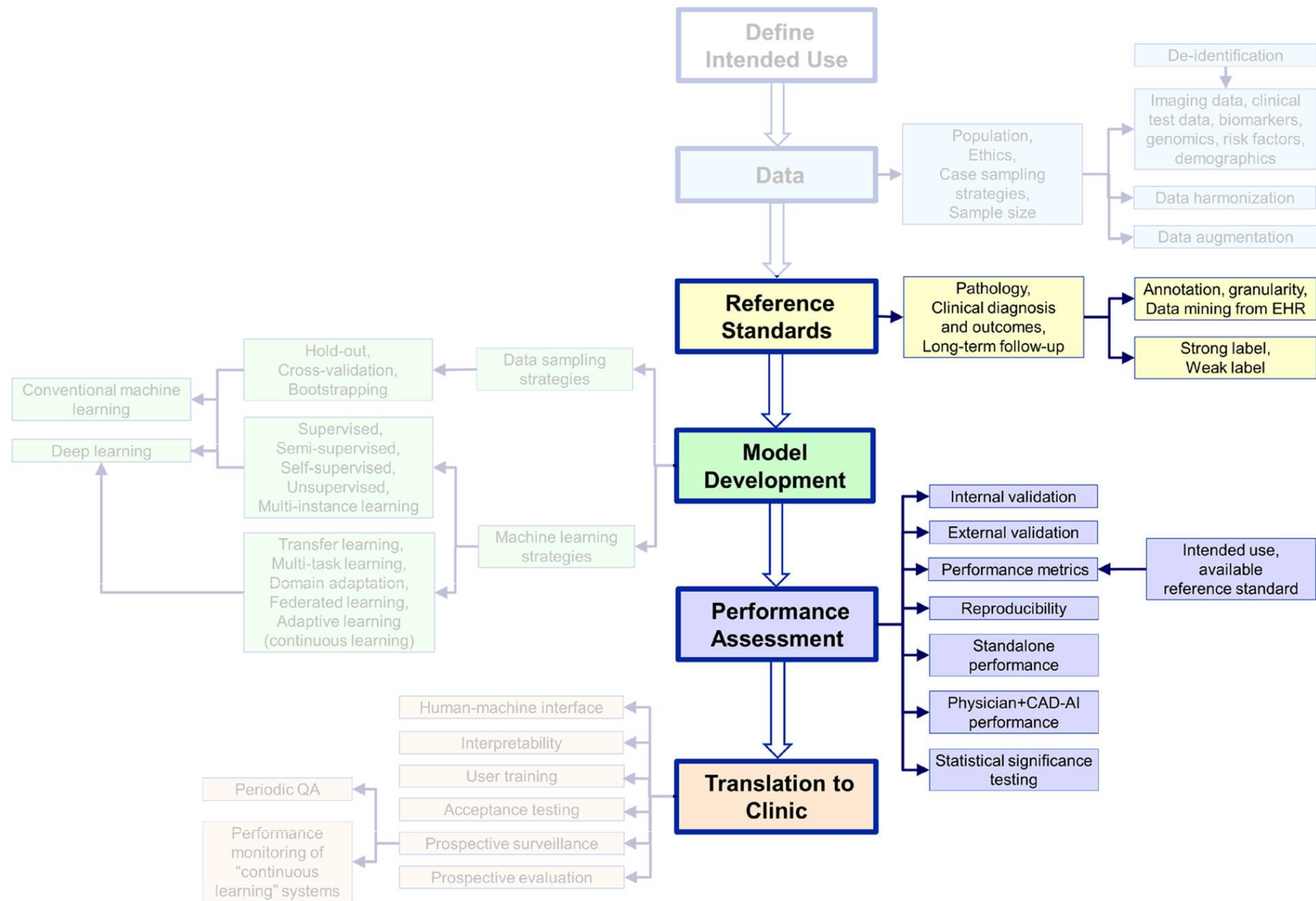
Source: <https://arxiv.org/pdf/1601.04674.pdf>



Source: Ruiwen Ding (UCLA)







Driving questions

1. How do we assess whether a model performs as intended?
2. How do we select between multiple models that accomplish the same task?
3. Will the model advance medical decision making?



Source: <https://xkcd.com/1838/>



Analytical Validation

- Does the learning algorithm correctly process input data to generate accurate, reliable, and precise output data?

Clinical Evaluation		
Valid Clinical Association	Analytical Validation	Clinical Validation
Is there a valid clinical association between your SaMD output and your SaMD's targeted clinical condition?	Does your SaMD correctly process input data to generate accurate, reliable, and precise output data?	Does use of your SaMD's accurate, reliable, and precise output data achieve your intended purpose in your target population in the context of clinical care?

Figure 1 - Clinical Evaluation Process

SaMD: Software as a Medical Device

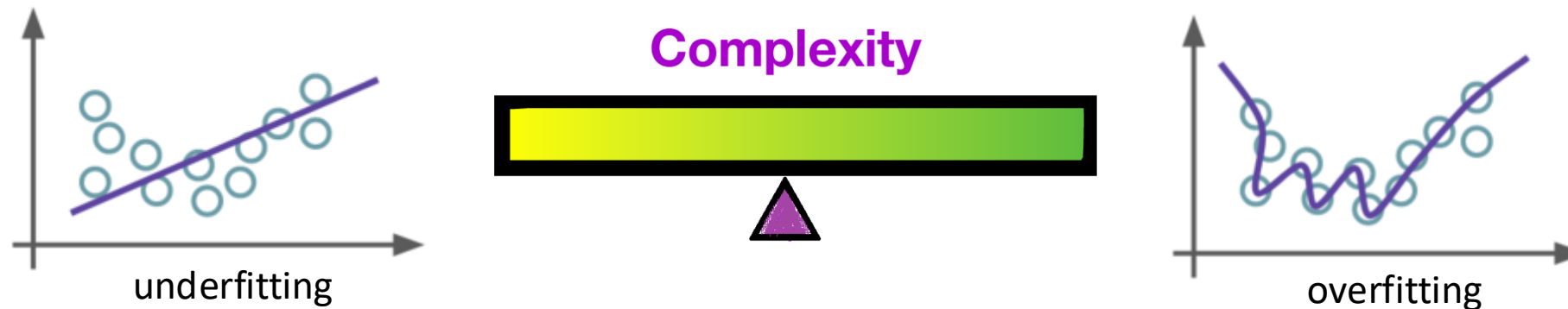
Source: <https://www.fda.gov/files/medical%20devices/published/Software-as-a-Medical-Device-%28SaMD%29--Clinical-Evaluation---Guidance-for-Industry-and-Food-and-Drug-Administration-Staff.pdf>

Basic concepts: Prediction error

- **Prediction error** can refer to one of two things:
 - In **regression analysis**, it is a measure of how well the model predicts the response variable
 - E.g., root mean squared error
 - In **classification**, it is a measure of how well samples are classified to the correct category
 - E.g., accuracy, precision, recall

Basic concepts: Bias-variance tradeoff

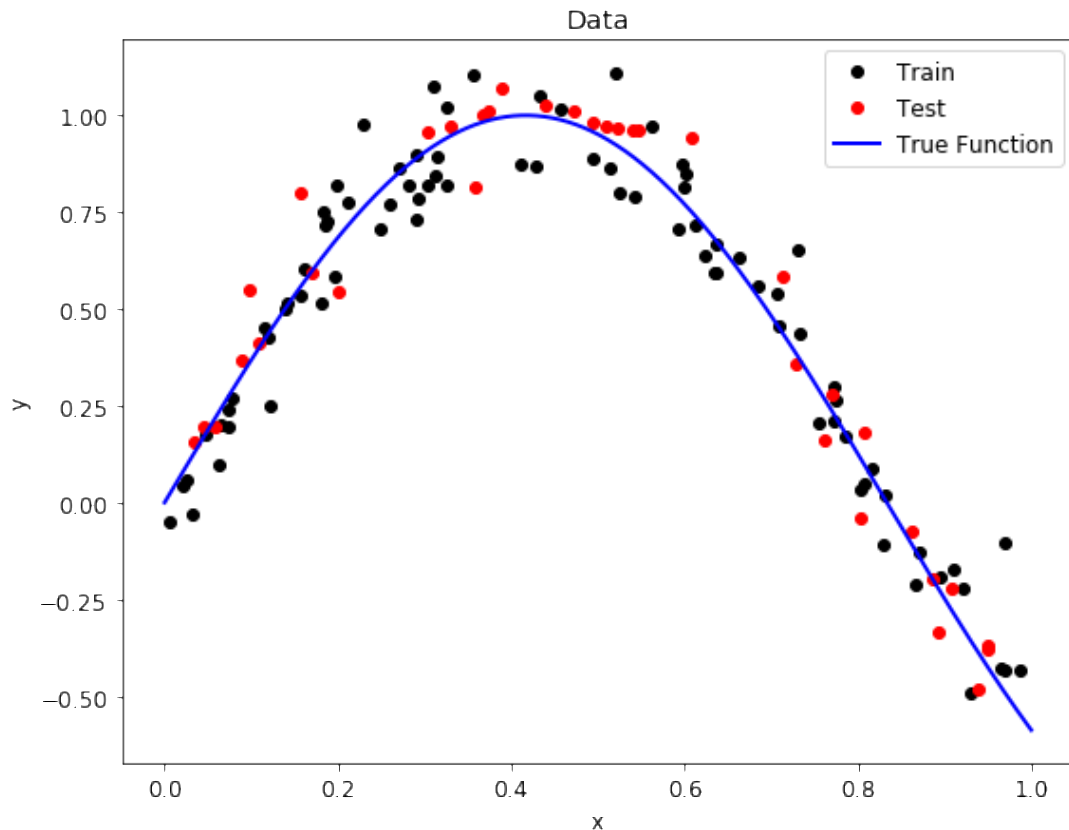
- **Bias** is the difference between the average prediction of our model and the correct value we are trying to predict.
- **Variance** is the model's sensitivity to small fluctuations in the training set. High variance can cause an algorithm to model the random noise in the training data rather than the intended outputs.



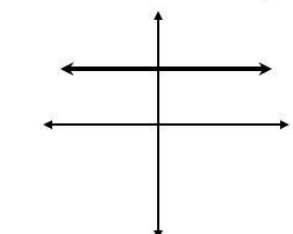
- *High bias & low variance = underfitting*
- *High variance & low bias = overfitting*

Regression example

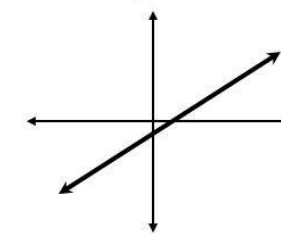
Evaluate polynomial function that best fits the data



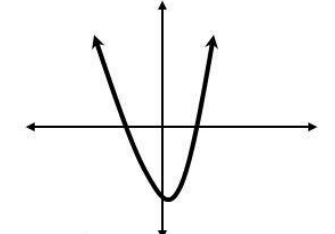
Graphs of Polynomial Functions:



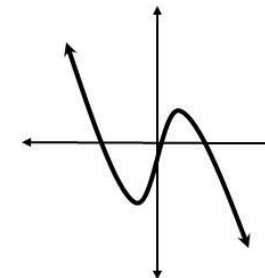
Constant Function
(degree = 0)



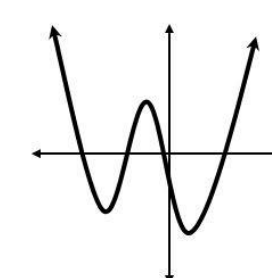
Linear Function
(degree = 1)



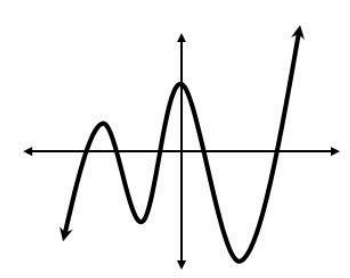
Quadratic Function
(degree = 2)



Cubic Function
(deg. = 3)



Quartic Function
(deg. = 4)

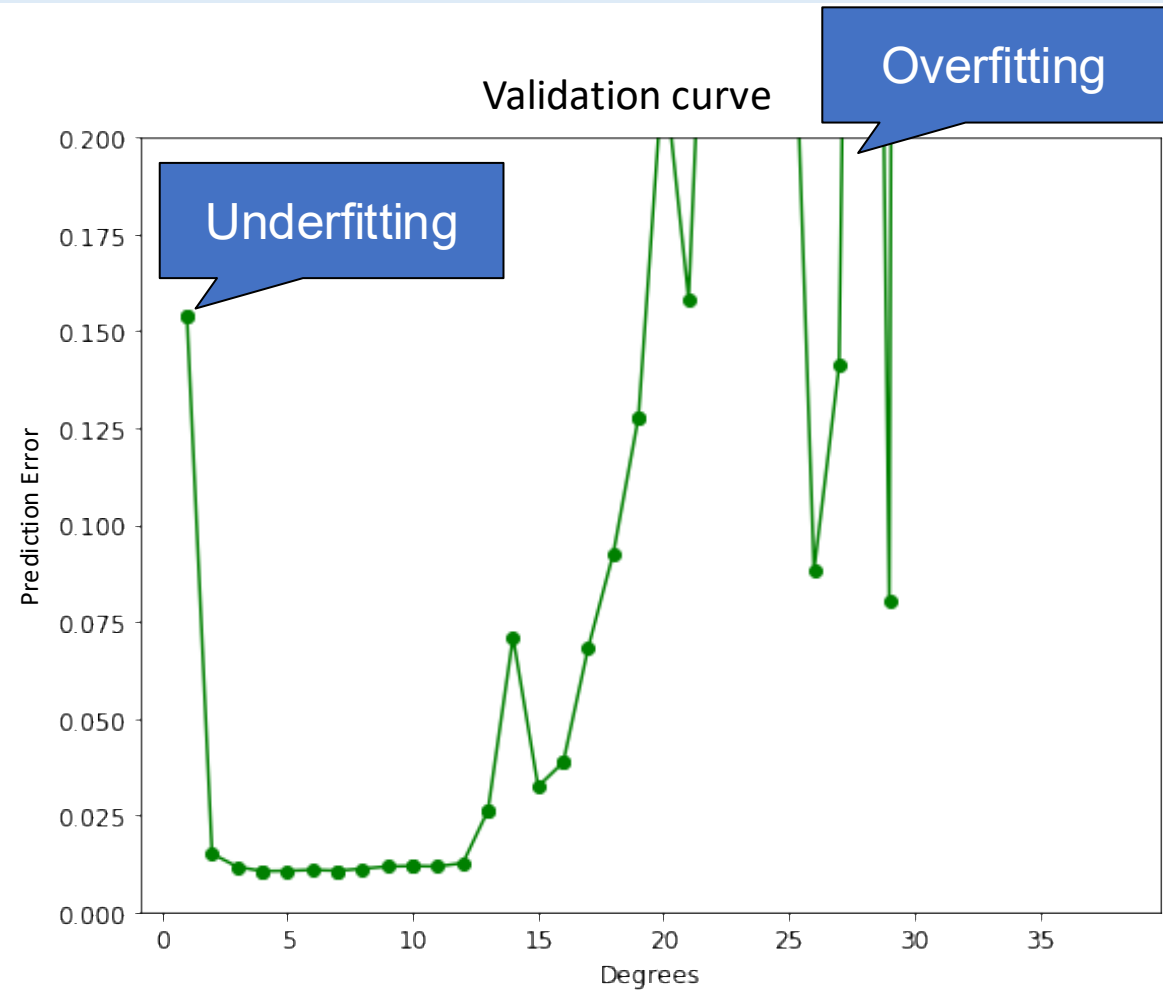


Quintic Function
(deg. = 5)

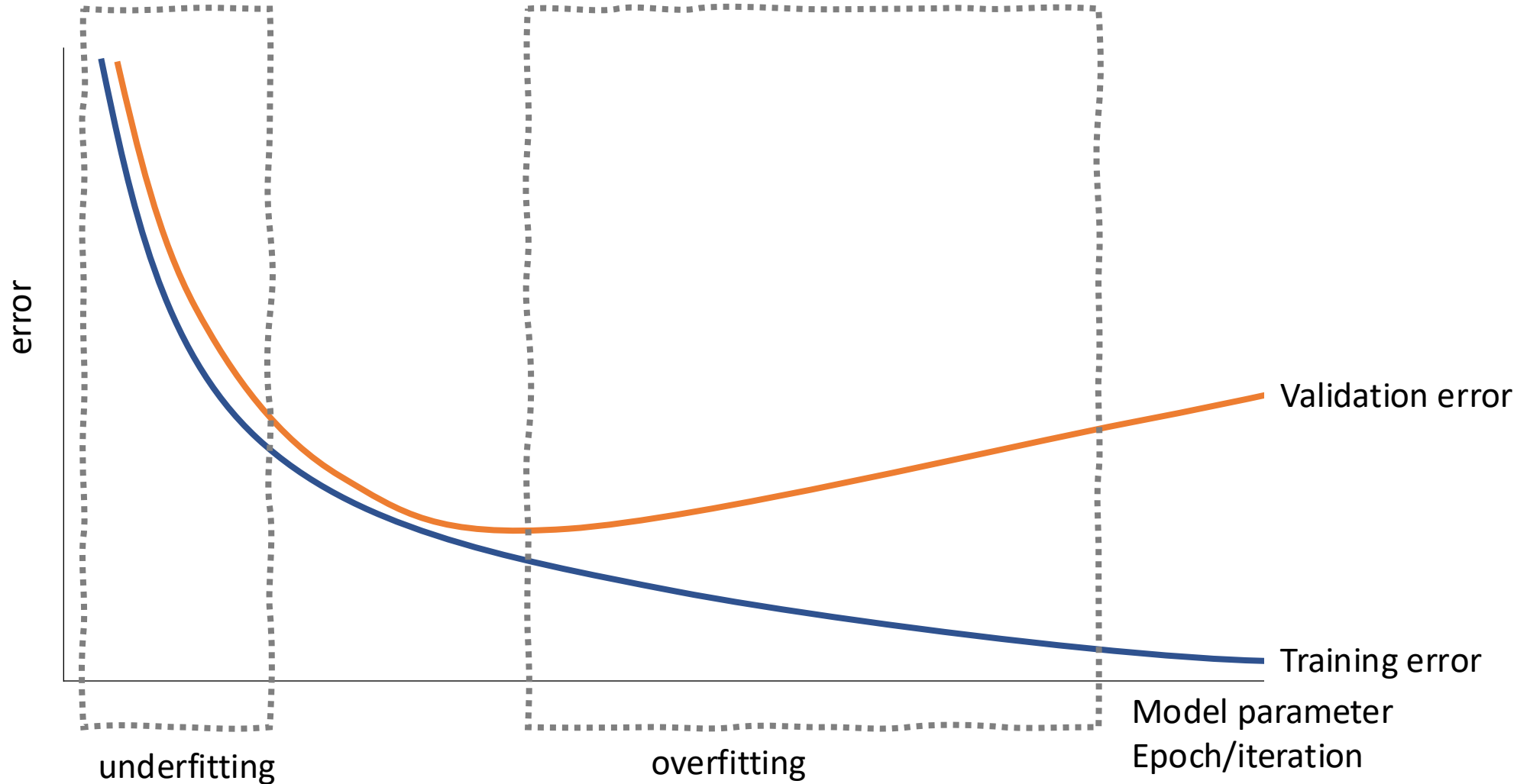
Regression example

Validation results
ordered by
prediction error

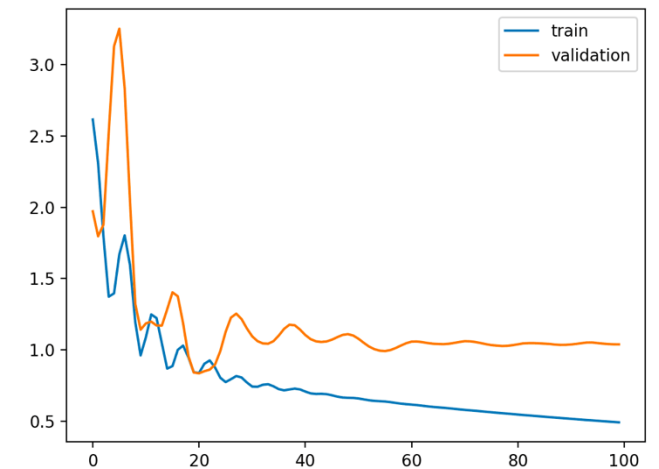
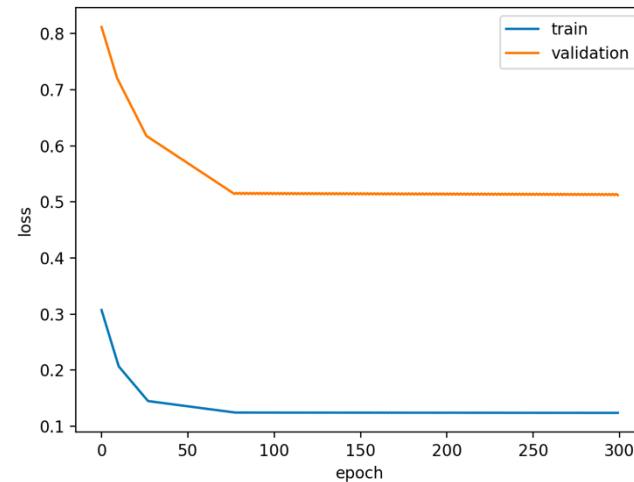
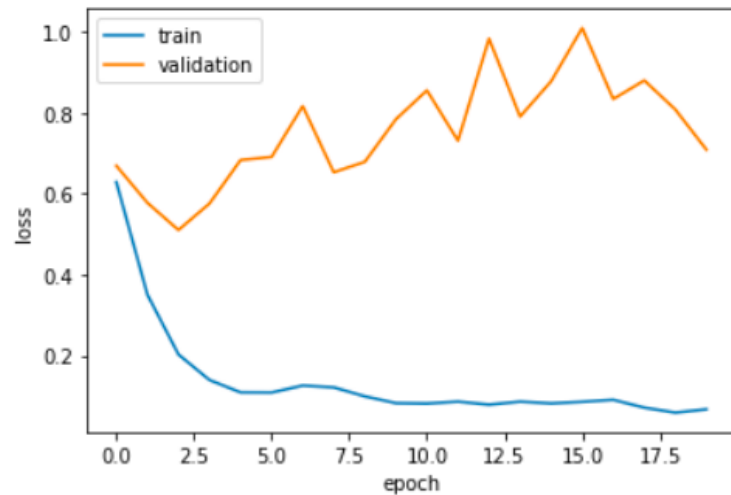
Degrees	MSE
4	0.010549
5	0.010637
7	0.010665
6	0.010887
8	0.011182
3	0.011695
9	0.011757
11	0.011769
10	0.011902
12	0.012642



Training and validation curves



Notes on training and validation curves



Confusion matrix

2-class problems

		Predicted Class	
		Positive	Negative
Real Class	Positive	True Positive (TP)	False Negative (FN)
	Negative	False Positive (FP)	True Negative (TN)

$$\text{accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{FP} + \text{FN} + \text{TN}}$$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{true positive rate (recall)} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{false positive rate (1 - specificity)} = \frac{\text{FP}}{\text{TN} + \text{FP}}$$

A single measure doesn't tell the whole story

Accuracy
(both 60%)

Precision =
66.7%

Recall = 40%

	Pos	Neg
Yes	200	100
No	300	400
	P=500	N=500

Precision =
57.1%

Recall = 80%

	Pos	Neg
Yes	400	300
No	100	200
	P=500	N=500

Precision/recall
(both 66.7% and 40%,
respectively)

True class →	Pos	Neg
Yes	200	100
No	300	400
	P=500	N=500

Accuracy = 60%

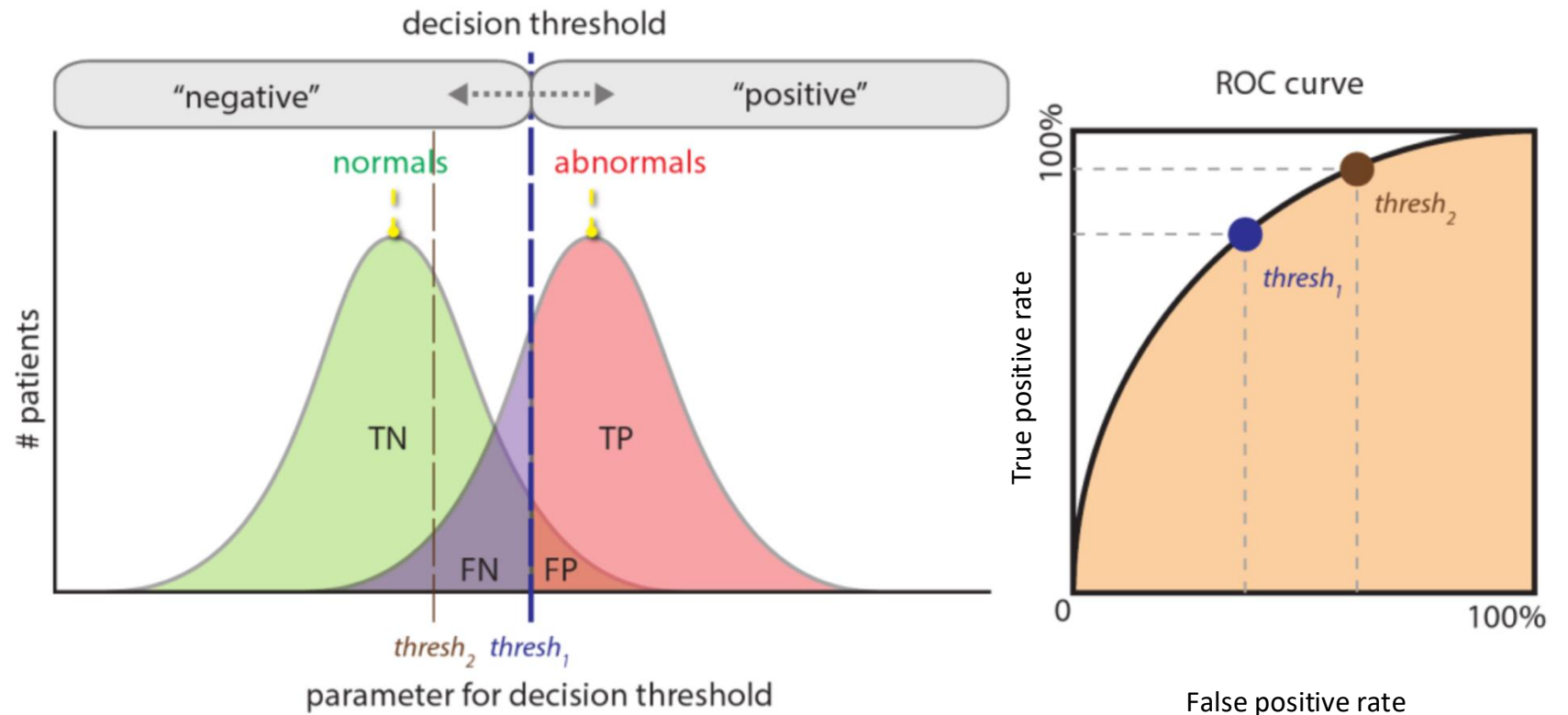
True class →	Pos	Neg
Yes	200	100
No	300	0
	P=500	N=100

Accuracy = 33%

Receiver operator characteristic (ROC) curve

$$\text{true positive rate (recall)} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

$$\text{false positive rate (1 - specificity)} = \frac{\text{FP}}{\text{TN} + \text{FP}}$$



ROC curve example

Benign=0; Malignant=1	Radiologist Likert scale (1-5) : 1 (~likely B) 5 (~likely M)	Threshold			
		≥ 2 (1/0)	≥ 3 (1/0)	≥ 4 (1/0)	$= 5$ (1/0)
0	1				
0	2				
0	3				
0	3				
0	4				
1	2				
1	3				
1	4				
1	4				
1	5				

Adapted from Grace H. Kim (UCLA)

Test 1, ≥ 2	Truth		
	+	-	Total
+ (≥ 2)			
- (1)			
Total	5	5	10

Sensitivity?
Specificity?

Test	Truth		
	+	-	Total
+ (≥ 3)			
- (1,2)			
Total	5	5	10

Sensitivity
Specificity

Test	Truth		
	+	-	Total
+ (≥ 4)			
- (≤ 3)			
Total	5	5	10

Sensitivity
Specificity

Test	Truth		
	+	-	Total
+ (5)			
- (≤ 4)			
Total	5	5	10

Sensitivity
Specificity

Benign=0; Malignant=1	Radiologist Likert scale (1-5) : 1 (~likely B) 5 (~likely M)	Threshold			
		≥2 (1/0)	≥3 (1/0)	≥4 (1/0)	≥5 (1/0)
0	1	0			
0	2	1			
0	3	1			
0	3	1			
0	4	1			
1	2	1			
1	3	1			
1	4	1			
1	4	1			
1	5	1			

Test 1, ≥2	Truth		
	+	–	Total
+ (≥2)	5	4	9
– (1)	0	1	1
Total	5	5	10

Sensitivity= 100%
Specificity= 20%
 1-specificity= 100-20=80%

Test	Truth		
	+	–	Total
+ (≥3)			
– (1,2)			
Total	5	5	10

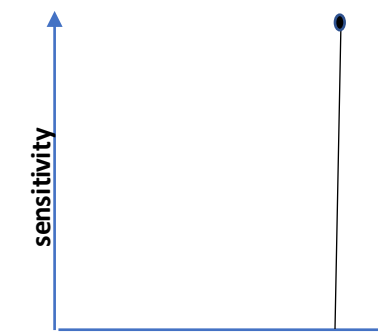
Sensitivity
Specificity

Test	Truth		
	+	–	Total
+ (≥4)			
– (≤3)			
Total	5	5	10

Sensitivity
Specificity

Test	Truth		
	+	–	Total
+ (5)			
– (≤4)			
Total	5	5	10

Sensitivity
Specificity



Area of trapezoid
 = 0.5(L1+L2)*Height

Benign=0; Malignant=1	Radiologist Likert scale (1-5) : 1 (~likely B) 5 (~likely M)	Threshold			
		≥2 (1/0)	≥3 (1/0)	≥4 (1/0)	=5 (1/0)
0	1	0	0		
0	2	1	0		
0	3	1	1		
0	3	1	1		
0	4	1	1		
1	2	1	0		
1	3	1	1		
1	4	1	1		
1	4	1	1		
1	5	1	1		

Test 1, ≥2	Truth		
	+	–	Total
+ (≥2)	5	4	9
– (1)	0	1	1
Total	5	5	10

Sensitivity=100%
Specificity=20%
 1-specificity=100-20=80%

Test	Truth		
	+	–	Total
+ (≥3)	4	3	7
– (1,2)	1	2	3
Total	5	5	10

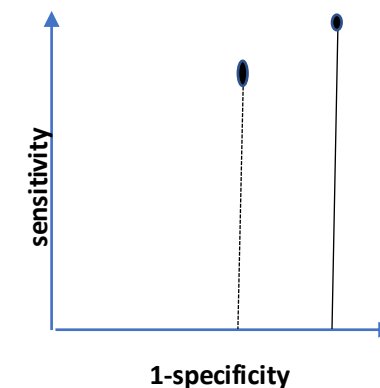
Sensitivity= 80%
Specificity =40%
 1-specificity=60%

Test	Truth		
	+	–	Total
+ (≥4)			
– (≤3)			
Total	5	5	10

Sensitivity
Specificity

Test	Truth		
	+	–	Total
+ (5)			
– (≤4)			
Total	5	5	10

Sensitivity
Specificity



Area of trapezoid
 = 0.5(L1+L2)*Height

Benign=0; Malignant=1	Radiologist Likert scale (1-5) : 1 (~likely B) 5 (~likely M)	Threshold			
		≥2 (1/0)	≥3 (1/0)	≥4 (1/0)	=5 (1/0)
0	1	0	0	0	0
0	2	1	0	0	0
0	3	1	1	0	0
0	3	1	1	0	0
0	4	1	1	1	0
1	2	1	0	0	0
1	3	1	1	0	0
1	4	1	1	1	0
1	4	1	1	1	0
1	5	1	1	1	1

Test 1, ≥2	Truth		
	+	–	Total
+ (≥2)	5	4	9
– (1)	0	1	1
Total	5	5	10

Sensitivity= 100%
Specificity= 20%
 1-specificity= 100-20=80%

Test	Truth		
	+	–	Total
+ (≥3)	4	3	7
– (1,2)	1	2	3
Total	5	5	10

Sensitivity= 80%
Specificity =40%
 1-specificity=60%

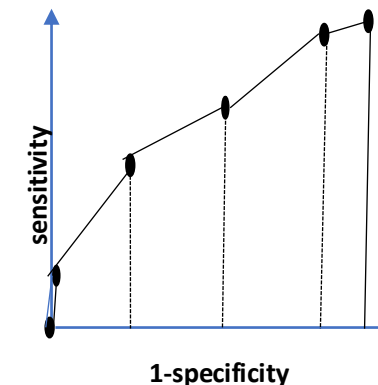
Test	Truth		
	+	–	Total
+ (≥4)	3	1	4
– (≤3)	2	4	6
Total	5	5	10

Sensitivity=60%
Specificity=80%
 1-specificity: 20%

Test	Truth		
	+	–	Total
+ (5)	1	0	1
– (≤4)	4	5	9
Total	5	5	10

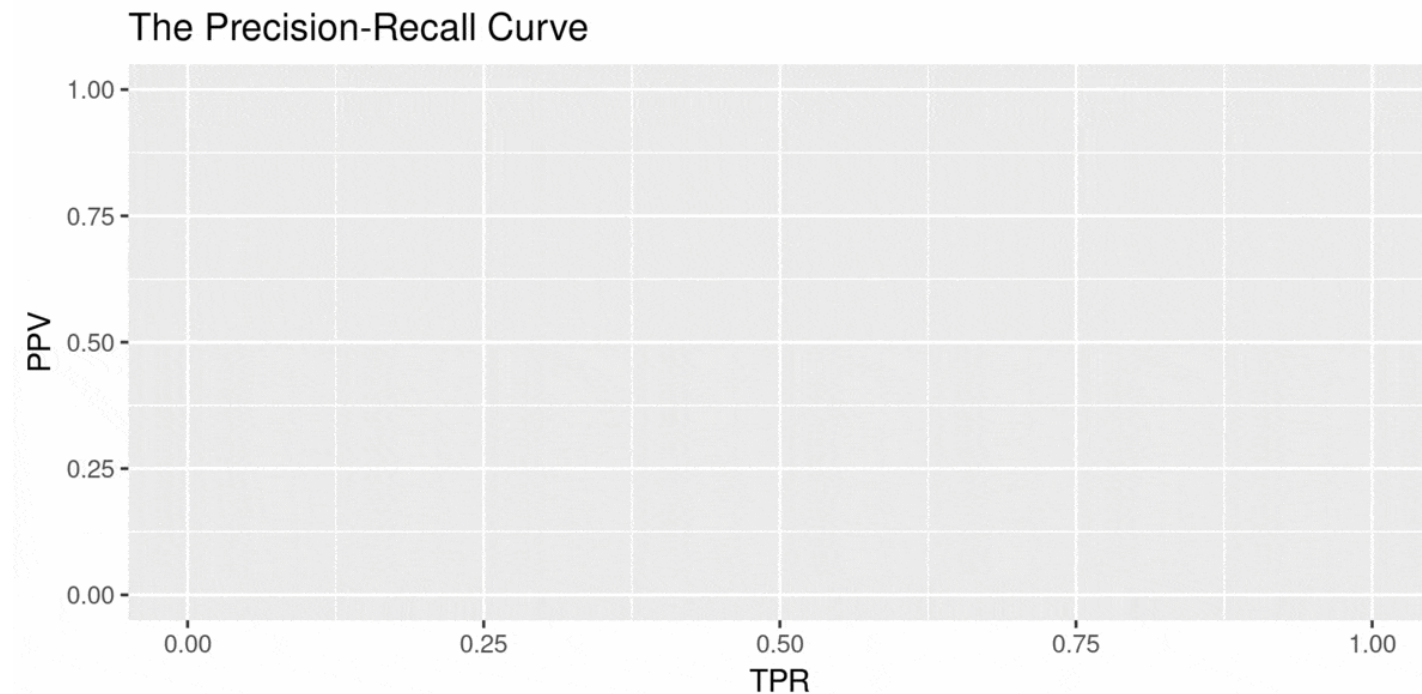
Sensitivity=20%
Specificity=100%
 1-specificity=0%

Area Under Curve(AUC)
 =Sum of trapezoids



AUC= Sum of trapezoids
 = 0.5[$((0.6+0.2)*0.2$
 $+ (0.6+0.8)*0.4$
 $+ (0.8+1)*0.2$
 $+ (1+1)*.2]$ =0.74

Precision-recall (PR) curves



Plots the precision vs. recall as a threshold on the confidence of an instance being positive is varied

<i>Estimate</i>	-3.0	-2.5	-1.0	-1.0	-0.5	-0.5	0.5	1.0	2.0	2.0	3.5
<i>Class</i>	-1	-1	-1	-1	-1	+1	-1	+1	-1	+1	+1
<i>Prediction</i>	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA
<i>Status</i>	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA	NA

Multi-class scenario

		Real Class		
		Class 1	Class 2	Class 3
Predicted Class	Class 1			
	Class 2			
	Class 3			

Multi-class scenario

		Actual		
		Benign	High risk	Malignant
Predicted	Benign	4	6	3
	High risk	1	2	0
	Malignant	1	2	6

Precision of 'benign' classifier
= $4/(4+6+3) = 30.8\%$

Recall of 'benign' classifier
= $4/(4+1+1) = 66.7\%$

Overall accuracy
= $(4+2+6)/25 = 48\%$

$$\text{precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad \text{true positive rate (recall)} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Multi-class scenario

- **No averaging:** Return a measure corresponding to each class.

		Actual		
		Benign	High risk	Malignant
Predicted	Benign	4	6	3
	High risk	1	2	0
	Malignant	1	2	6

- Precision of 'benign' classifier = $4/(4+6+3) = 30.8\%$
- Precision of 'high risk' classifier = $2/(1+2) = 66.7\%$
- Precision of 'malignant' classifier = $6/(1+2+6) = 66.7\%$

Multi-class scenario

- **Macro averaging:**
Calculate the measure independently for each class then find their unweighted mean.

		Actual		
		Benign	High risk	Malignant
Predicted	Benign	4	6	3
	High risk	1	2	0
	Malignant	1	2	6

- Macro-precision of classifier is
 $(31\% + 67\% + 67\%) / 3 = 54.7\%$

Multi-class scenario

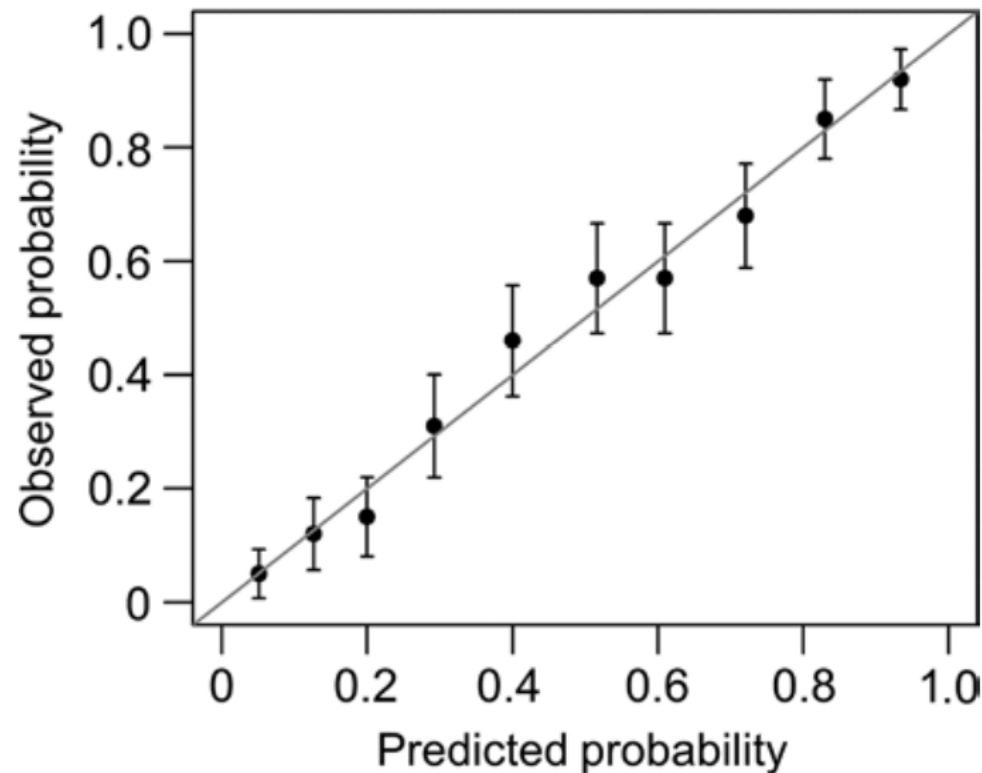
- **Micro averaging:**
Aggregate contributions of all classes to compute the average measure by counting the total number of times each class was correctly predicted and incorrectly predicted.

		Actual		
		Benign	High risk	Malignant
Predicted	Benign	4	6	3
	High risk	1	2	0
	Malignant	1	2	6

- $TP=4+2+6=12$
- $FP=6+3+1+0+1+2=13$
- Micro-precision of classifier is $12/(12+13) = 48\%$

Calibration

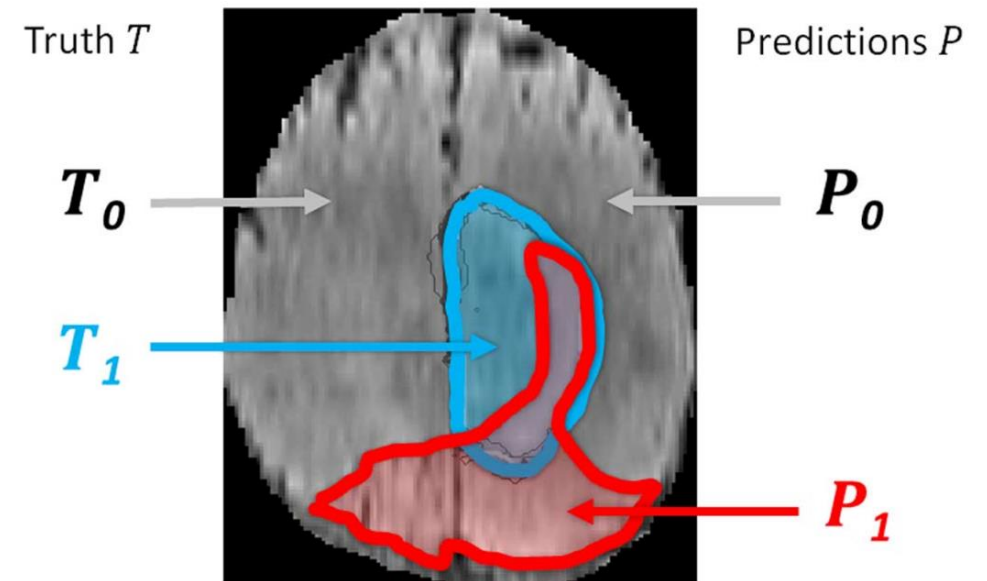
- How well does the model's predicted probability approximates the actual event probability



Other measures of interest

- Comparing segmented regions
 - Overlap measures
 - Dice coefficient
 - Intersection over union (Jaccard)
 - Surface distance
 - Hausdorff distance
- Evaluating clustering algorithms
 - Purity
 - Rand index
 - <http://nlp.stanford.edu/IR-book/html/htmledition/evaluation-of-clustering-1.html>

$$\text{Dice}(P, T) = \frac{|P_1 \wedge T_1|}{(|P_1| + |T_1|)/2}$$



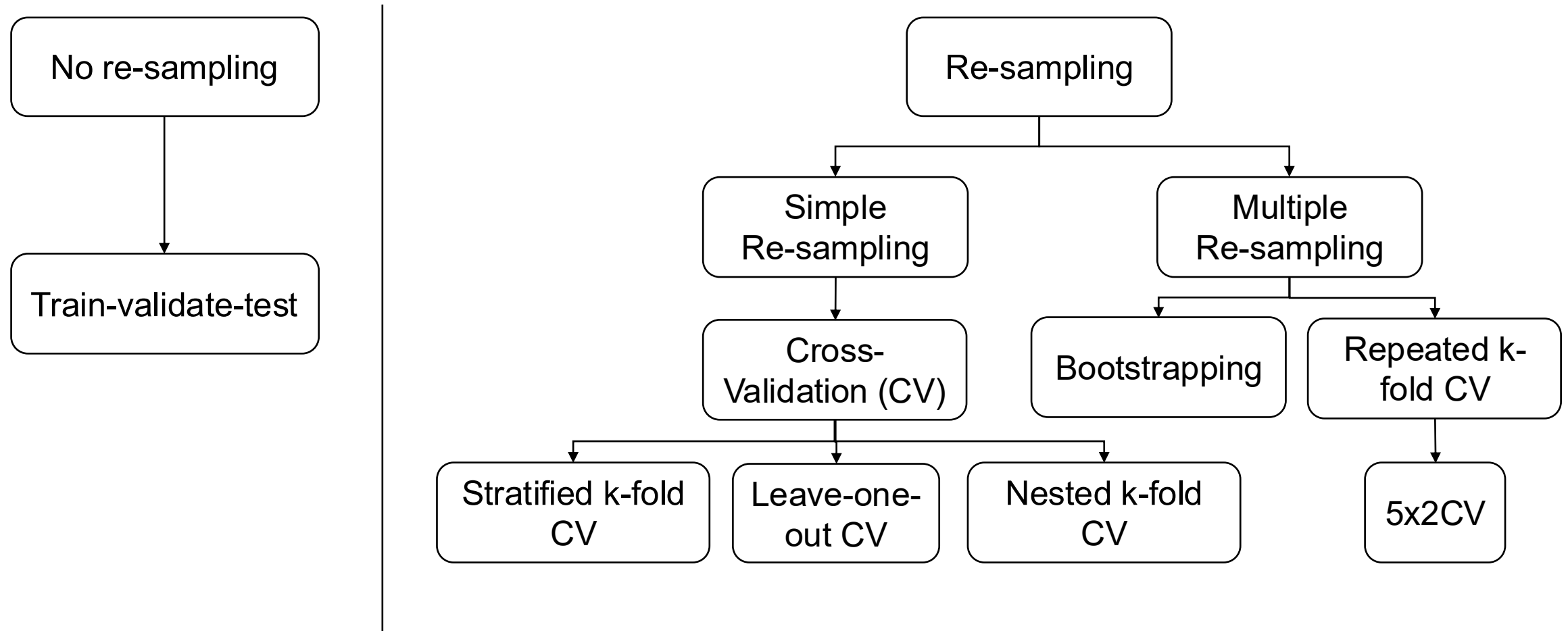
Notes on measures

Regression	Classification	Segmentation	Image generation	Text summarization
Means squared error	Accuracy	Intersection over union	Peak signal-to-noise ratio (PSNR)	Bilingual evaluation understudy (BLEU)
Mean absolute error	Precision-recall curve	Dice coefficient	Structural similarity index (SSIM)	Recall-Oriented Understudy for Gisting Evaluation (ROUGE)
Root mean squared error	F1-score	Hausdorff distance	Fréchet Inception Distance (FID)	
R^2	Receiver operating characteristic curve			

Notes on measures

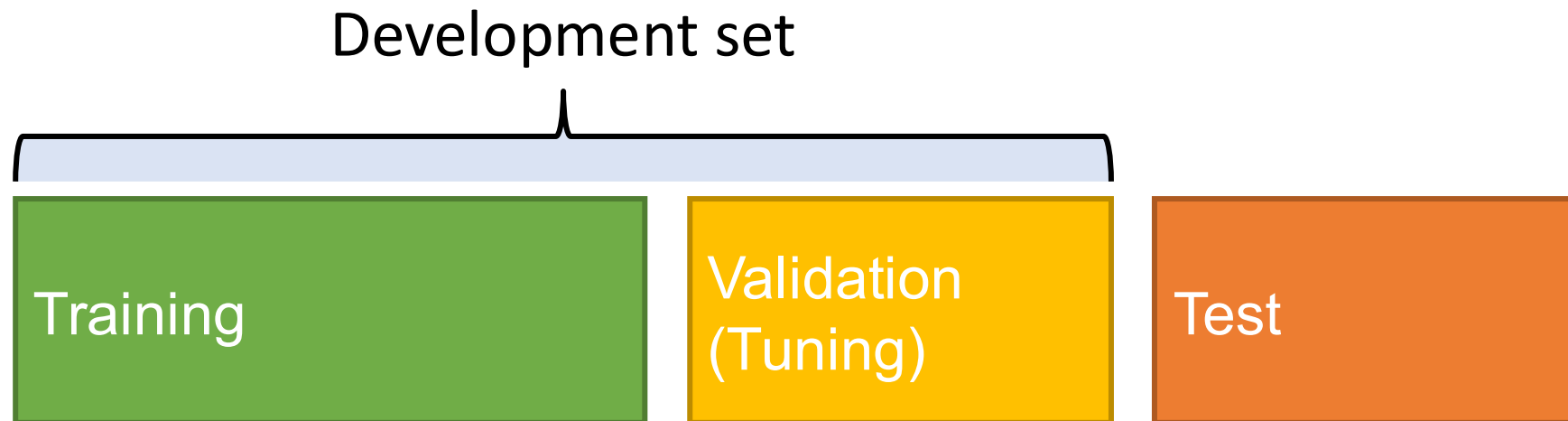
- Choice of measure is based on type of outcome variable and task
 - What errors matter the most?
- Variants:
 - F1 (equal weighting of precision-recall) vs F2 score (place more emphasis on recall)
 - Free-response ROC “FROC” (average number of false positives per image vs true positive localization fraction)
- Class imbalance: ROC-AUC (insensitive to prevalence) vs PR curve (focuses on minority class)

Evaluation designs



Train-validation-test

- Split data into three groups
- Use training data to fit different models
- Use validation data to estimate generalization error (and tune model)
- Use test data to assess the performance

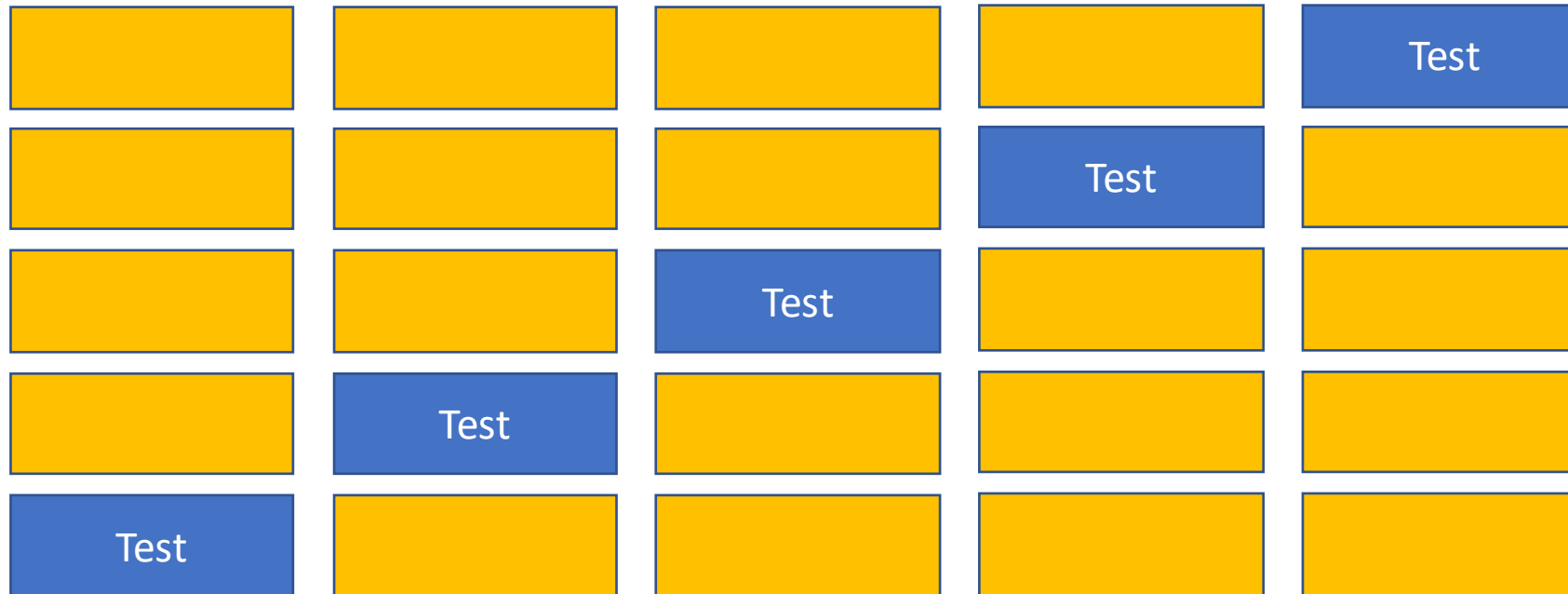


Notes on train-validation-test

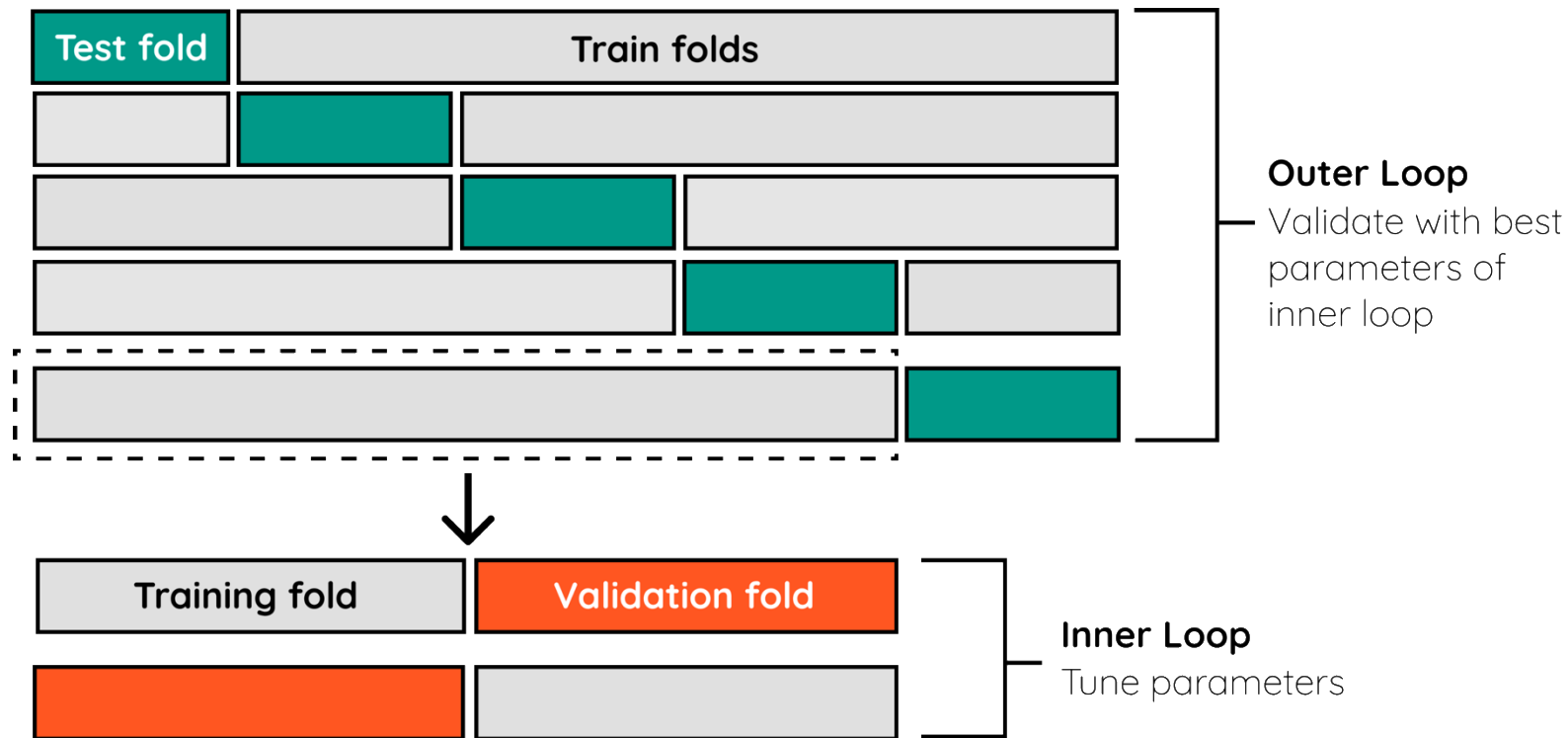
- Often need to balance between number of training and testing cases
 - Larger training set provides the learning algorithm with a more representative sample of the data
 - Larger test sets provide a more reliable estimate of accuracy (lower variance estimate)
- A single training set does not reveal how sensitive the accuracy of the learning algorithm is to a particular training sample → sampling bias

Cross validation

- Partition data into k-subsamples then iteratively assign one subsample as the test set, train the learning algorithm on the rest

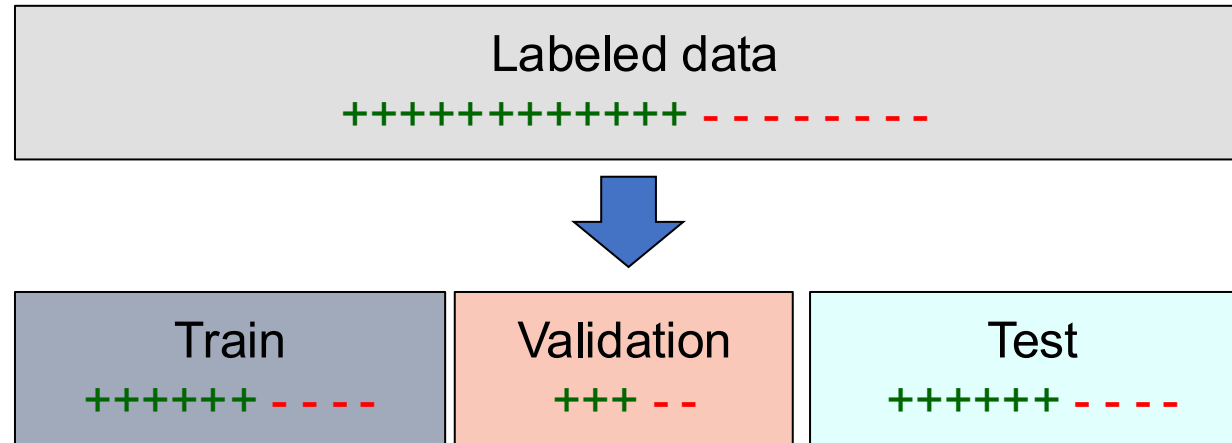


Nested cross validation



Reference: <https://medium.com/data-science/validating-your-machine-learning-model-25b4c8643fb7>

Stratified cross-validation



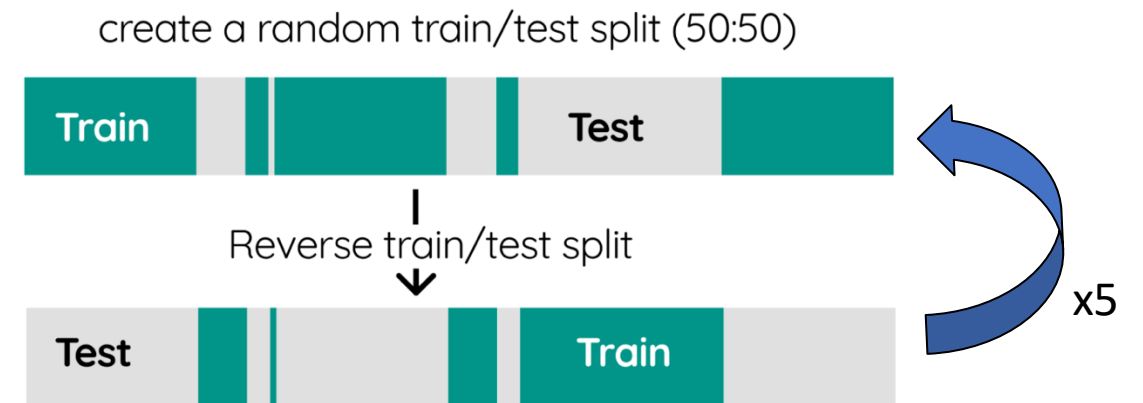
When randomly selecting training or validation sets, stratified cross-validation is used to ensure that class proportions are maintained in each selected set

Bootstrapping

- Bootstrapping assumes that available samples are representative and creates a large number of new samples by drawing with replacement from the available samples
- Bootstrapping can be useful when the sample size is too small for cross validation or leave-one-out approaches to yield a good estimate

Repeated k-fold cross-validation

- To obtain more stable estimates of an algorithm's performance, perform multiple runs of simple re-sampling schemes
 - Dietterich (1998) proposed the 5x2CV, in which a two-fold cross-validation is repeated 5 times



Notes about evaluation design

- Goal of cross-validation studies: identify the optimal modeling approach, not an individual learned model
 - Use cross validation to determine optimal modeling approach
 - Train model using optimal approach on the entire development set
 - Evaluate final model on the held-out test set
- 10-fold cross-validation is commonly used, but smaller values of k can be used when learning takes a lot of time
- When working with time series data, training data must always be selected before the test data (otherwise training data will contain information from the future)

Comparing models: Motivation

- Which model is better?

Algo	Acc	RMSE	TPR	FPR	Prec	Rec	F	AUC	Info S
NB	71.7	.4534	.44	.16	.53	.44	.48	.7	48.11
C4.5	75.5	.4324	.27	.04	.74	.27	.4	.59	34.28
3NN	72.4	.5101	.32	.1	.56	.32	.41	.63	43.37
Ripp	71	.4494	.37	.14	.52	.37	.43	.6	22.34
SVM	69.6	.5515	.33	.15	.48	.33	.39	.59	54.89
Bagg	67.8	.4518	.17	.1	.4	.17	.23	.63	11.30
Boost	70.3	.4329	.42	.18	.5	.42	.46	.7	34.48
RanF	69.23	.47	.33	.15	.48	.33	.39	.63	20.78

What is being tested?

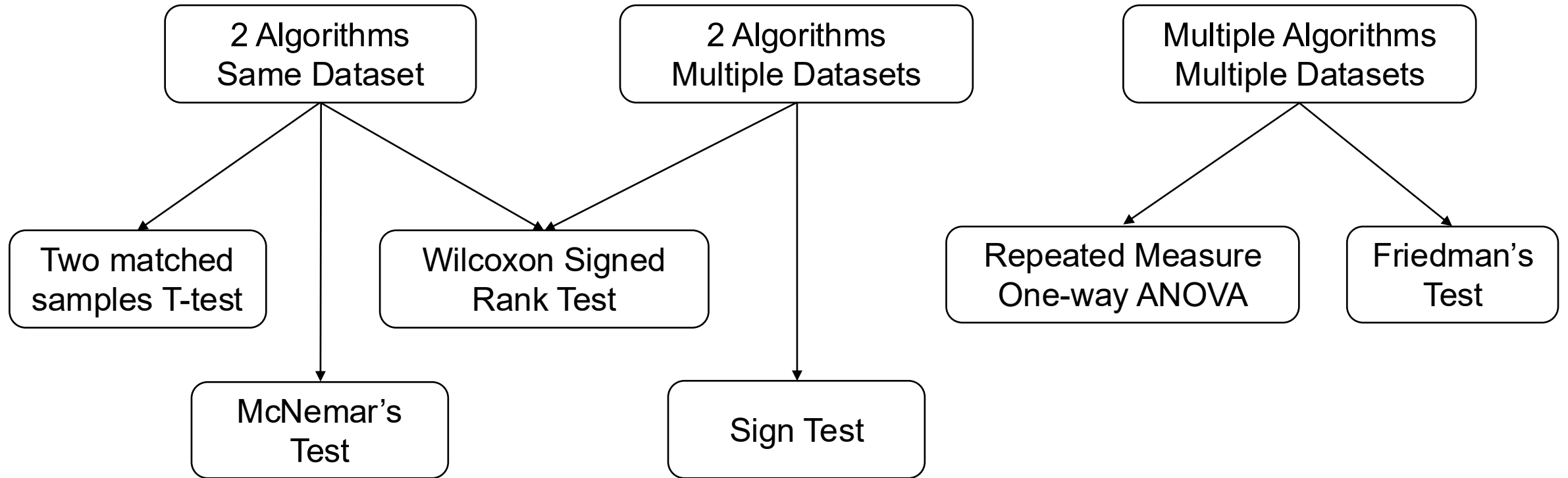
Are the performance measurements attributable to the machine learning algorithm or are they observed by chance?

- Provide evidence that our selected evaluation measure is representative of the general behavior of our machine learning algorithm
- Parametric versus nonparametric test
 - Parametric tests make strong assumptions about the distribution of the underlying data (normal, Bernoulli, etc) but require smaller sample size
 - Nonparametric tests make weaker assumptions (skewed distribution, presence of outliers) but are also less powerful (less apt at rejecting the null hypothesis when it is false)

Considerations

- Can a parametric versus nonparametric test be used
- Select appropriate test based on the following scenarios
 - The comparison of 2 algorithms on the same dataset
 - The comparison of 2 algorithms on different datasets
 - The comparison of multiple algorithms on multiple datasets

Test selection



The paired t-test

- Given two matched samples
 - the results of two classifiers applied to the same data set with **matching randomizations** and **partitions**
- Test whether the difference in **mean of a metric (e.g., accuracy)** between these two classifiers is significant
 - Test whether the two samples come from the same population.
 - Look at the distribution in **observed means** and **standard deviation**.
- We assume that the difference between means is zero (the null hypothesis) and see if we can reject this hypothesis.

Notes on using the t-test

- **Normality:** The t-test requires that the samples come from normally distributed population.
- **Randomness:** The sample should be representative of the underlying population.
- **Equal variance:** The two samples come from populations with equal variance.
- Measures effect, but not the size of the effect → Pearson correlation coefficient

	Classifier A	Classifier B	Difference	
Iteration 1	P_{A1}	P_{B1}	$d_1 = P_{A1} - P_{B1}$	P_{Ai} : Performance of classifier A at iteration “i” P_{Bi} : Performance of classifier B at iteration “i”
Iteration 2	P_{A2}	P_{B2}	$d_2 = P_{A2} - P_{B2}$	
Iteration 3	P_{A3}	P_{B3}	$d_3 = P_{A3} - P_{B3}$	
Iteration 4	P_{A4}	P_{B4}	$d_4 = P_{A4} - P_{B4}$	
	$\vdots$	$\vdots$	$\vdots$	
Iteration n-1	$P_{A(n-1)}$	$P_{B(n-1)}$	$d_{n-1} = P_{A(n-1)} - P_{B(n-1)}$	$\bar{d} = \frac{1}{n} \sum_{i=1}^n d_i$ $\bar{\sigma}^2 = \frac{\sum_{i=1}^n (d_i - \bar{d})^2}{n - 1}$
Iteration n	P_{An}	P_{Bn}	$d_n = P_{Bn} - P_{Bn}$	

McNemar's test

- Non-parametric counterpart of the t-test observed on the test set
 - # of instances misclassified by both classifiers C_{00}
 - # of instances misclassified by first classifier but correctly classified by the second classifier C_{01}
 - # of instances misclassified by the second classifier but correctly classified by the first classifier C_{10}
 - # of instances correctly classified by both classifiers C_{11}

		Classifier f_2	
		0	1
Classifier f_1	0	c_{00}^{Mc}	c_{01}^{Mc}
	1	c_{10}^{Mc}	c_{11}^{Mc}

$$X_{MC}^2 = \frac{(|C_{01} - C_{10}| - 1)^2}{C_{01} + C_{10}}$$

Notes on using the McNemar's test

- Similar to a t-test on dichotomous rather than continuous values
- Does not require the same assumptions about the distribution as the t-test
- Applied over an independent test set
- $C_{01} + C_{10}$ cannot be too “small” (<25 discordant pairs) → does not follow X^2 distribution and must use a binomial distribution instead (sign test)

The Sign test

- The sign test can be used either to compare
 - two classifiers on a single domain (using the results at each fold as a trial)
 - two classifiers on multiple domains
- We count
 - number of times that algorithm 1 outperforms algorithm 2, n_{a1}
 - number of times that algorithm 2 outperforms algorithm 1, n_{a2}
- The null hypothesis (stating that the two classifiers perform equally well) holds if the number of wins follows a binomial distribution.
- Practically speaking, a classifier should perform better on at least w_α datasets to be considered statistically significantly better at the α significance level, where w_α is the critical value for the sign test at the α significance level

Notes on using the sign test

Dataset	NB	SVM	Adaboost	Rand Forest
Anneal	96.43	99.44	83.63	99.55
Audiology	73.42	81.34	46.46	79.15
Balance Scale	72.30	91.51	72.31	80.97
Breast Cancer	71.70	66.16	70.28	69.99
Contact Lenses	71.67	71.67	71.67	71.67
Pima Diabetes	74.36	77.08	74.35	74.88
Glass	70.63	62.21	44.91	79.87
Hepatitis	83.21	80.63	82.54	84.58
Hypothyroid	98.22	93.58	93.21	99.39
Tic-Tac-Toe	69.62	99.90	72.54	93.94

- SVM $\leq$ NB: $n_{f1} = 4.5$, $n_{f2} = 5.5$ and $w_{0.05} = 8 \rightarrow$ we cannot reject the null hypothesis stating that NB and SVM perform similarly on these data sets for level $\alpha=0.05$ (1-tailed)
- RF $\leq$ ADA: $n_{f1} = 1.5$ and $n_{f2} = 8.5 \rightarrow$ we can reject the null hypothesis at level $\alpha=0.05$ (1-tailed) and conclude that RF is significantly better than Ada on these data sets at that significance level.

Notes on using the sign test

A.4 The Table of Critical Values for the Signed Test

Table A.4. *Critical values of T for the sign test*

Level of significance α					Level of significance α				
Two-sided	0.10	0.05	0.02	0.01	Two-sided	0.10	0.05	0.02	0.01
One-sided	0.05	0.025	0.01	0.005	One-sided	0.05	0.025	0.01	0.005
n					n				
1	–	–	–	–	31	11	13	15	17
2	–	–	–	–	32	12	14	16	16
3	–	–	–	–	33	11	13	15	17
4	–	–	–	–	34	12	14	16	16
5	5	–	–	–	35	11	13	15	17
6	6	6	–	–	36	12	14	16	18
7	7	7	7	–	37	11	13	17	17
8	6	8	8	8	38	12	14	16	18
9	7	7	9	9	39	13	15	17	17
10	8	8	10	10	40	12	14	16	18
11	7	9	9	11	45	13	15	17	19
12	8	8	10	10	46	14	16	18	20
13	7	9	11	11	49	13	15	19	19
14	8	10	10	12	50	14	16	18	20

Wilcoxon's Signed-Rank Test

- Non-parametric
- For each domain, we calculate the difference in the performance of the two classifiers.
 - Rank the absolute values of these differences and graft the signs in front of the ranks.
 - Calculate the sum of positive and negative ranks, respectively (W_{S1} and W_{S2})
 - $T_{Wilcox} = \min(W_{S1}, W_{S2})$
- Compare to critical value V_{α} . If $V_{\alpha} \geq T_{Wilcox}$ we reject the null hypothesis that the performance of the two classifiers is the same, at the α confidence level.

Wilcoxon's signed-rank test: Illustration

Data	NB	SVM	NB-SVM	NB-SVM	Ranks	$\pm$ Ranks
1	.9643	.9944	-0.0301	0.0301	3	-3
2	.7342	.8134	-0.0792	0.0792	6	-6
3	.7230	.9151	-0.1921	0.1921	8	-8
4	.7170	.6616	+0.0554	0.0554	5	+5
5	.7167	.7167	0	0	Remove	Remove
6	.7436	.7708	-0.0272	0.0272	2	-2
7	.7063	.6221	+0.0842	0.0842	7	+7
8	.8321	.8063	+0.0258	0.0258	1	+1
9	.9822	.9358	+0.0464	0.0464	4	+4
10	.6962	.9990	-0.3028	0.3028	9	-9

$H_0: \text{SVM} \leq \text{NB}$

$W_{S1} = 17$ and $W_{S2} = 28 \rightarrow T_{\text{Wilcox}} = \min(17, 28) = 17$

For $n = 10 - 1$ degrees of freedom and $\alpha = 0.005$, $V = 8$ for the 1-sided test. V must be larger than T_{Wilcox} in order to reject the hypothesis. Since $17 > 8$, we cannot reject the null hypothesis

Analysis of variance (ANOVA)

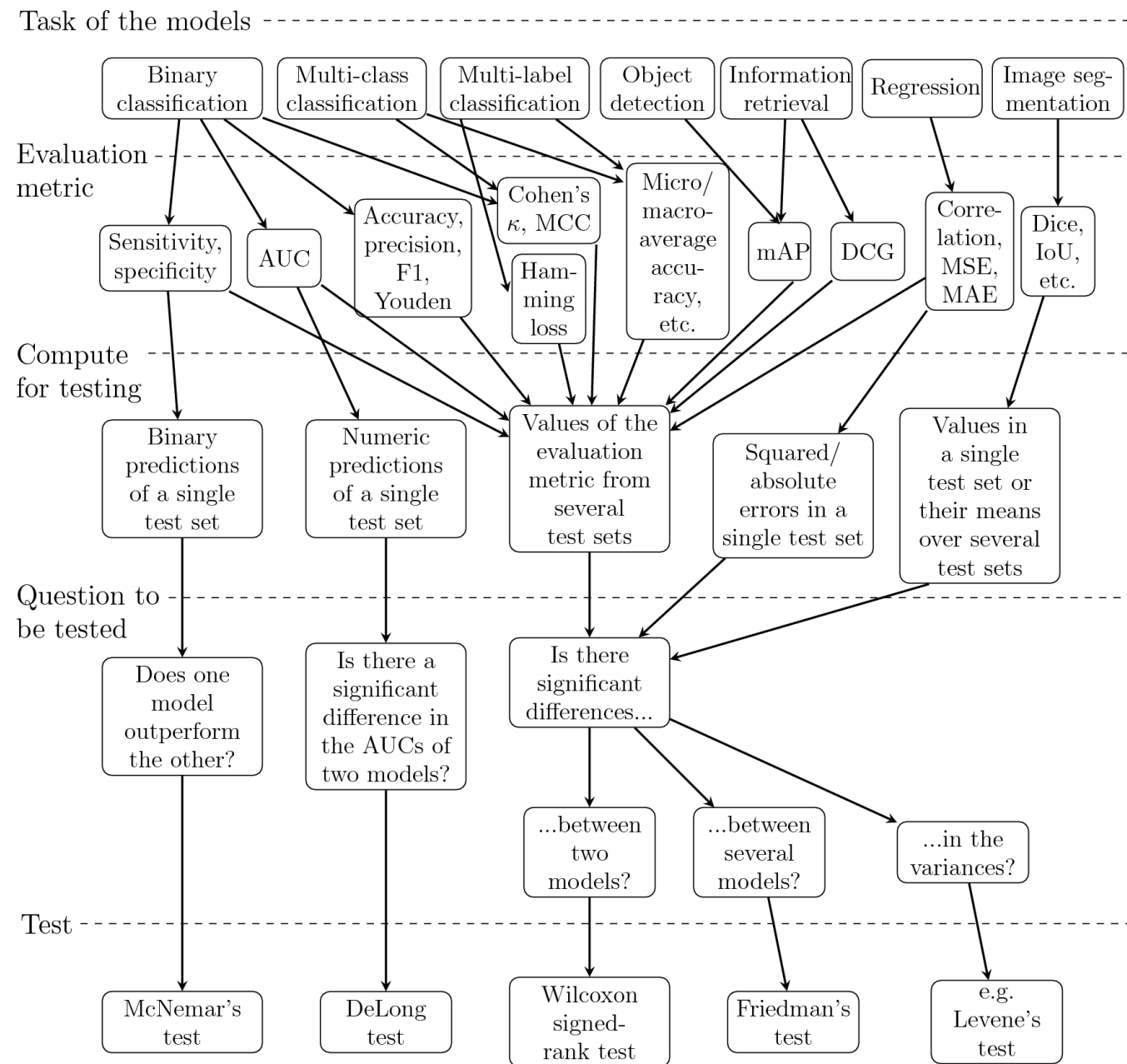
- For the case of multiple classifiers and multiple domains, two alternatives are possible
 - The parametric alternative is (one-way repeated measure) ANOVA
 - The non-parametric alternative is Friedman's Test
- H_0 : all the classifiers perform equally
- H_A : at least one pair of classifiers with significantly different performance
- If H_0 is rejected, the test is followed by a post-hoc test whose job is to identify the significantly different pairs of classifiers

ANOVA post-hoc tests

Approach	Tukey's HSD Procedure	Scheffé's Procedure	Newman–Keuls Procedure	Dunnett's Procedure	Games-Howell procedure
Assumption	Equal variance assumed, equal sample size NOT assumed	Equal variance assumed, equal sample size NOT assumed	Equal variance and sample size assumed	Equal variance NOT assumed, equal sample size NOT assumed	Equal variance NOT assumed, equal sample size NOT assumed
Details	- Only pairwise comparisons - Has a multiplier based on the number of treatment levels and the degrees of freedom for error mean square	- All possible simple and complex pairs of comparisons (most versatile) - E.g., compare the overall mean of two or more dosage level with a placebo - A higher critical value is used to determine significance (most conservative)	- Only pairwise comparisons - Depends on the steps that separate pairs of means - Less conservative than Tukey's test - Can't form confidence intervals for mean differences	- Only in situations in which several treatment means are compared with a single control mean - Can't compare between treatment means - Relatively low critical value	- When statistical power is to be considered, this test consistently provides narrower confidence intervals - Robust to non normality
Notes	- Equal variances - Only pairwise comparisons	- Equal variances - All possible comparisons	- Equal variances & equal sample sizes - If unequal sample sizes, need adjustment	- Unequal variance & sample size - Only: several treatment means are compared with a single control mean	Unequal variance & sample size

Notes about statistical testing

- What are we asked to compare?
 - Mean(s), variance(s), proportion(s)
 - One-sample, two-sample, etc.
- What statistical test to use?
 - Assumptions
 - Normally distributed? Equal variances?
- How to perform the test?
 - (Assumptions)
 - 1. Null and alternative hypotheses
 - One-sided or two-sided
 - 2. Alpha level
 - 3. Test statistic
 - Manually
 - Software
 - 4. Statistical decision
 - Manually: compare test statistic to critical value
 - Software: compare the p-value to the alpha level
 - 5. Conclusion



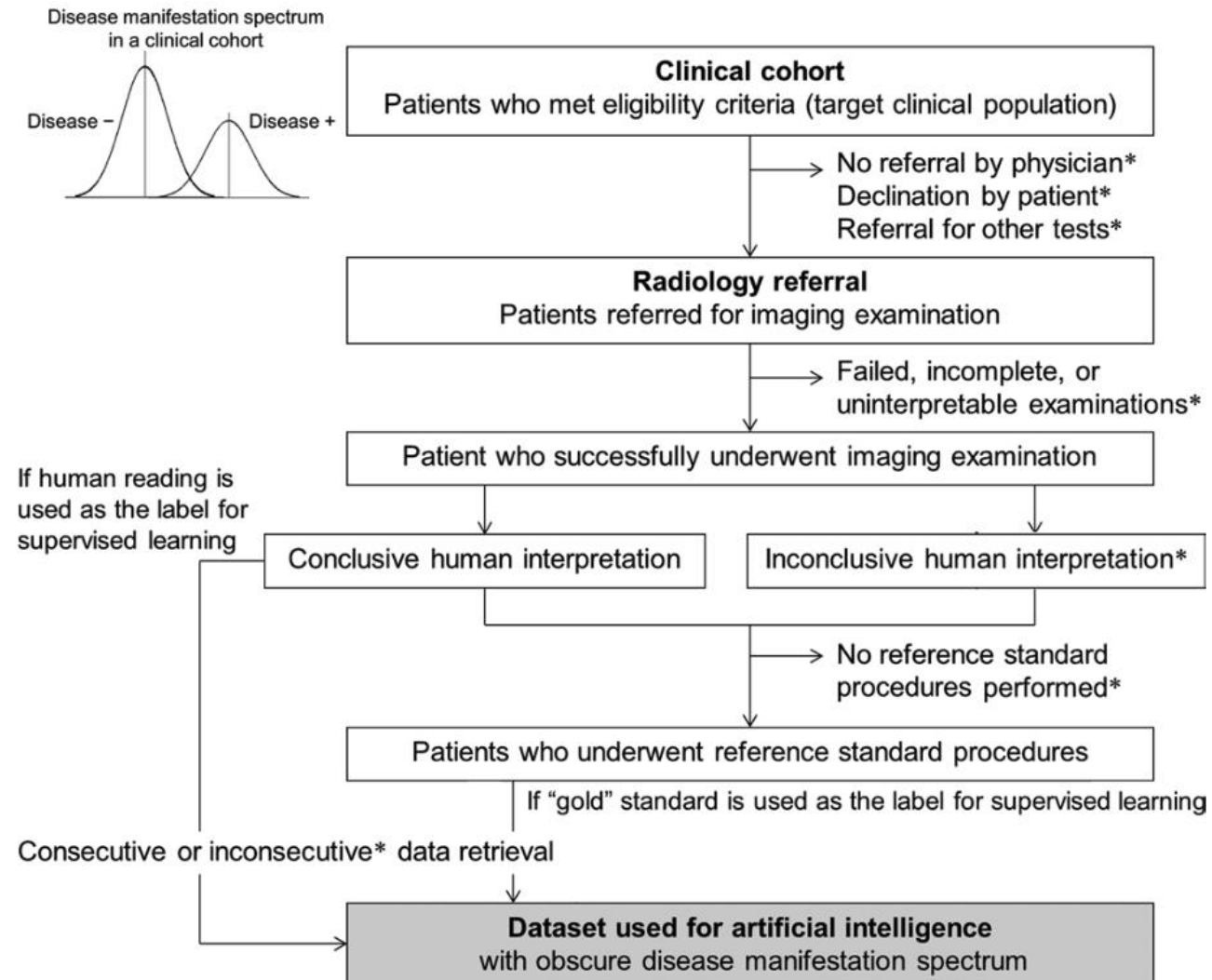
Notes about statistical testing

Statistical significance $\neq$ Clinical significance

Guidelines for reporting data

- Data sources
 - Inclusion criteria
 - Selection of data subsets
- Data pre-processing steps
 - Definition of data elements
 - De-identification methods
 - How missing data were handled
- Reference standard
 - How was the standard defined?
 - The rationale for choosing the standard
 - Source of ground truth annotations
 - If generated by human annotators, were inter and intra-rater variability mitigated?

Data sources



Reference standard

- Source of the standard
 - Completeness?
 - Consistent?
- Rationale for choosing the standard
- How was the standard defined?
 - Subject to measurement error?
 - Subjective judgment?

	Majority Decision of Retinal Specialist Grading before Adjudication				
	No	Mild	Moderate	Severe	Proliferative
Adjudicated Consensus					
No	1469	4	5	0	0
Mild	58	62	5	0	0
Moderate	22	3	118	1	0
Severe	0	0	13	36	1
Proliferative	0	0	0	1	15

Confusion matrix for diabetic retinopathy between the grade determined by majority decision and adjudicated consensus.

Source: <https://doi.org/10.1016/j.ophtha.2018.01.034>

Guidelines for reporting models

- Model
 - Description of model
 - Inputs and outputs
 - Hyperparameters
- Implementation
 - Software libraries and frameworks
 - Computational requirements
- Training
 - How was the dataset split for training?
 - Was transfer learning used?
 - Method of selecting final model

Guidelines for reporting evaluation

- Evaluation approach
 - Metrics used to measure model performance
 - Statistical tests used
 - Whether sensitivity analysis was performed to evaluate model robustness
 - Techniques for explaining model results, if applicable
 - Validation of model using external data
- Results
 - Whether performance metrics were compared with previously reported models, if possible
 - Whether confidence intervals are provided
 - Failure analysis
- Limitations
 - Possible sources of bias
 - Implications for practice

Helpful reporting standards

- Checklist for Artificial Intelligence in Medical Imaging (CLAIM)
 - <https://pubs.rsna.org/page/ai/claim>
- Transparent reporting of a multivariable prediction model for individual prognosis or diagnosis (TRIPOD)
 - <https://www.equator-network.org/reporting-guidelines/tripod-statement/>
 - TRIPOD-AI statement: <https://www.bmj.com/content/385/bmj-2023-078378>
- Prediction model Risk Of Bias ASsessment Tool (PROBAST)
 - <https://www.acpjournals.org/doi/10.7326/M18-1376>
 - PROBAST-AI toolkit: <https://www.bmj.com/content/388/bmj-2024-082505>
- Reporting of clinical trials involving AI
 - CONSORT/CONSORT-AI: <https://www.nature.com/articles/s41591-020-1034-x>
 - SPIRIT (protocol reporting): <https://www.nature.com/articles/s41591-020-1037-7>

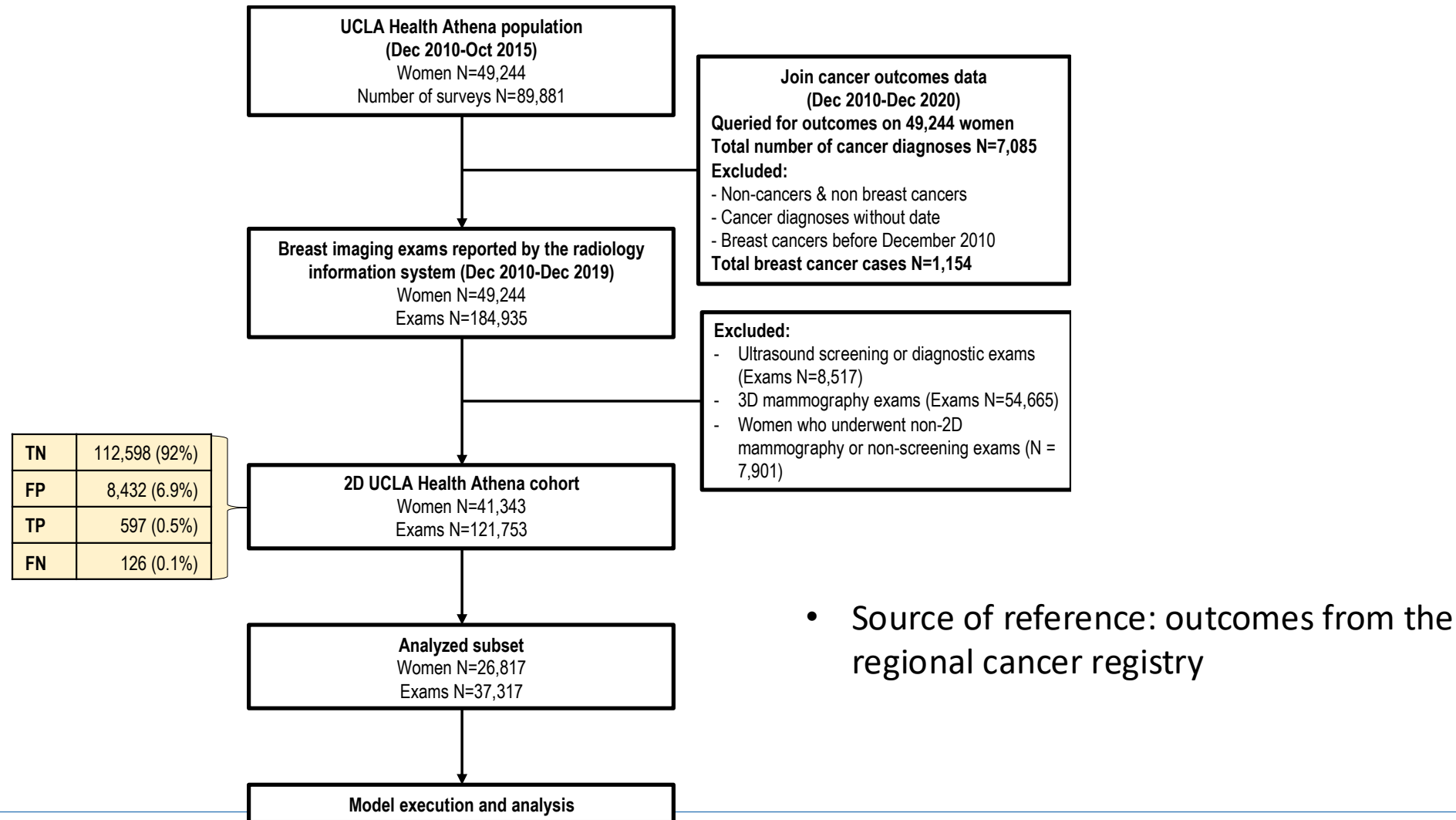
Example: External validation

- Obtain a suitable dataset
 - Appropriateness of the data sources and inclusion/exclusion criteria used
 - Predictors defined/assessed all participants in a similar way?
 - Predictor assessments made without knowledge of the outcome?
 - Predictors available at the time the model is intended to be used?
 - Outcome pre-specified, consistently defined, and determined appropriately?
- Generate predictions from the model
- Quantify model performance
- Provide clear and transparent reporting

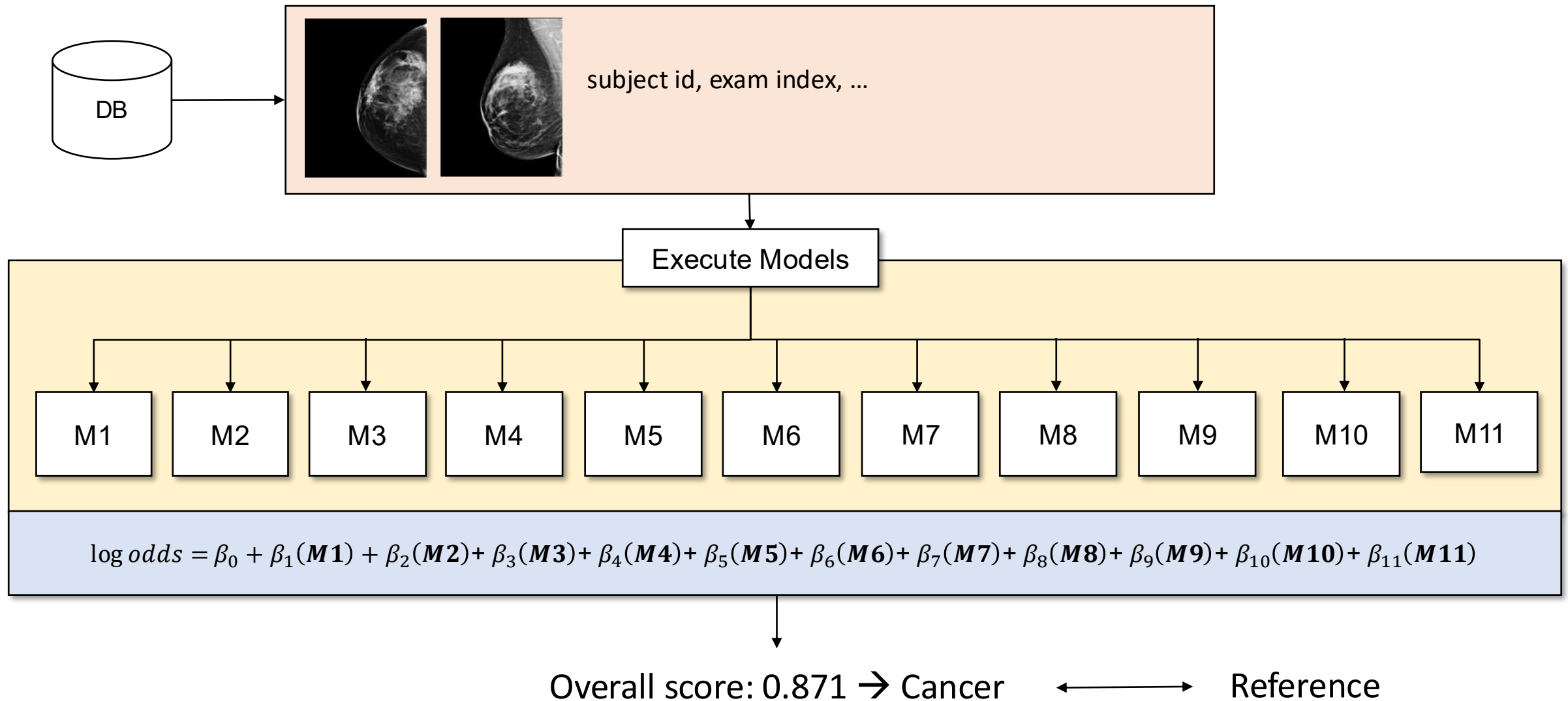
Example: External validation

- New breast AI/ML algorithms are becoming commercially available and are being reported to have performance on par with breast radiologists
 - Lack of transparency about the cohorts in which the algorithm was trained and evaluated
→ will the performance generalize?
 - Guidelines on how institutions should review and adopt AI/ML algorithms are sparse
→ what factors should institutions consider before buying?
 - Check if the AI/ML algorithm works as intended
→ when does an algorithm deviate from stated performance?
- Objective: Perform an external validation of the "DREAM Challenge Competition Ensemble Method" model on a retrospective cohort from our institution
 - Primary measures: AUC, sensitivity, specificity
 - Identify conditions when model performance differs from previously reported results

Example: External validation

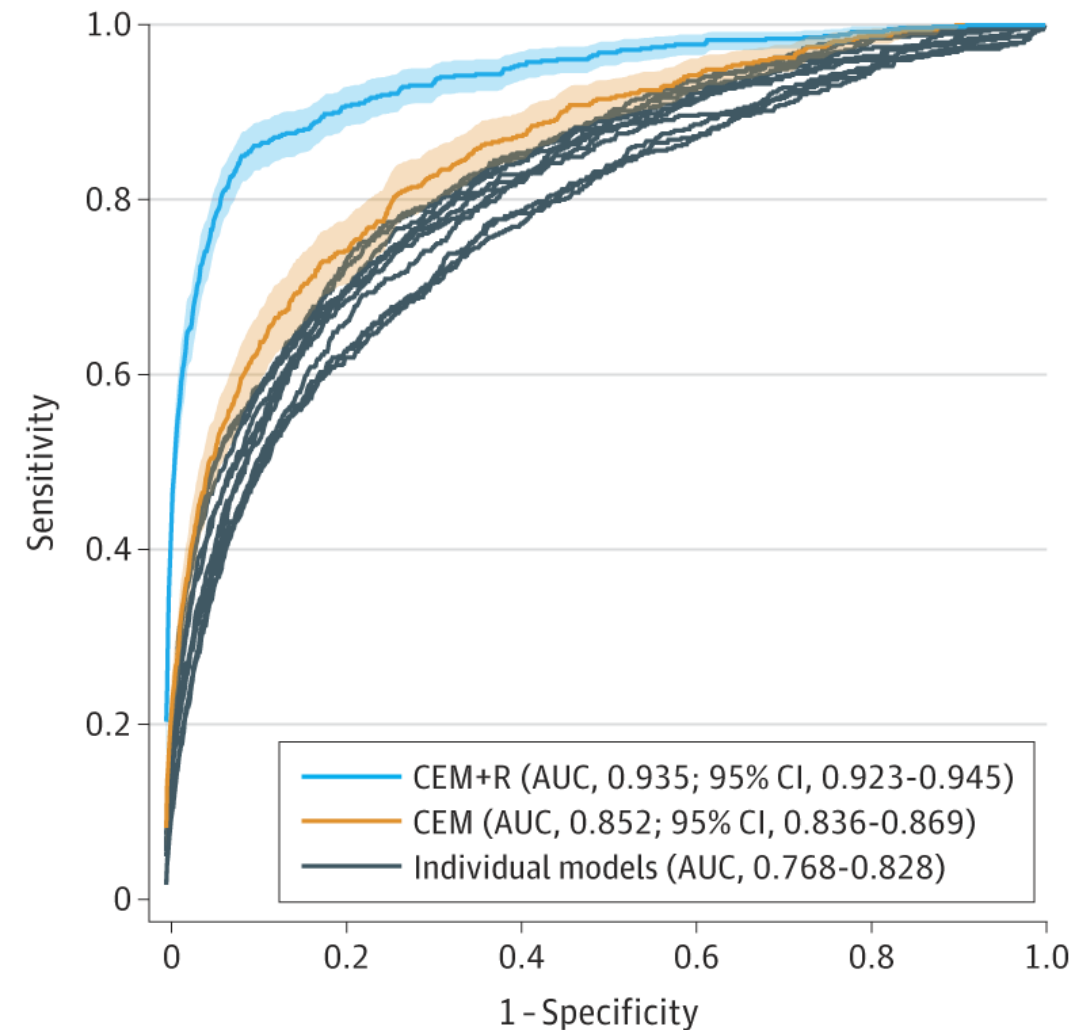


Example: External validation

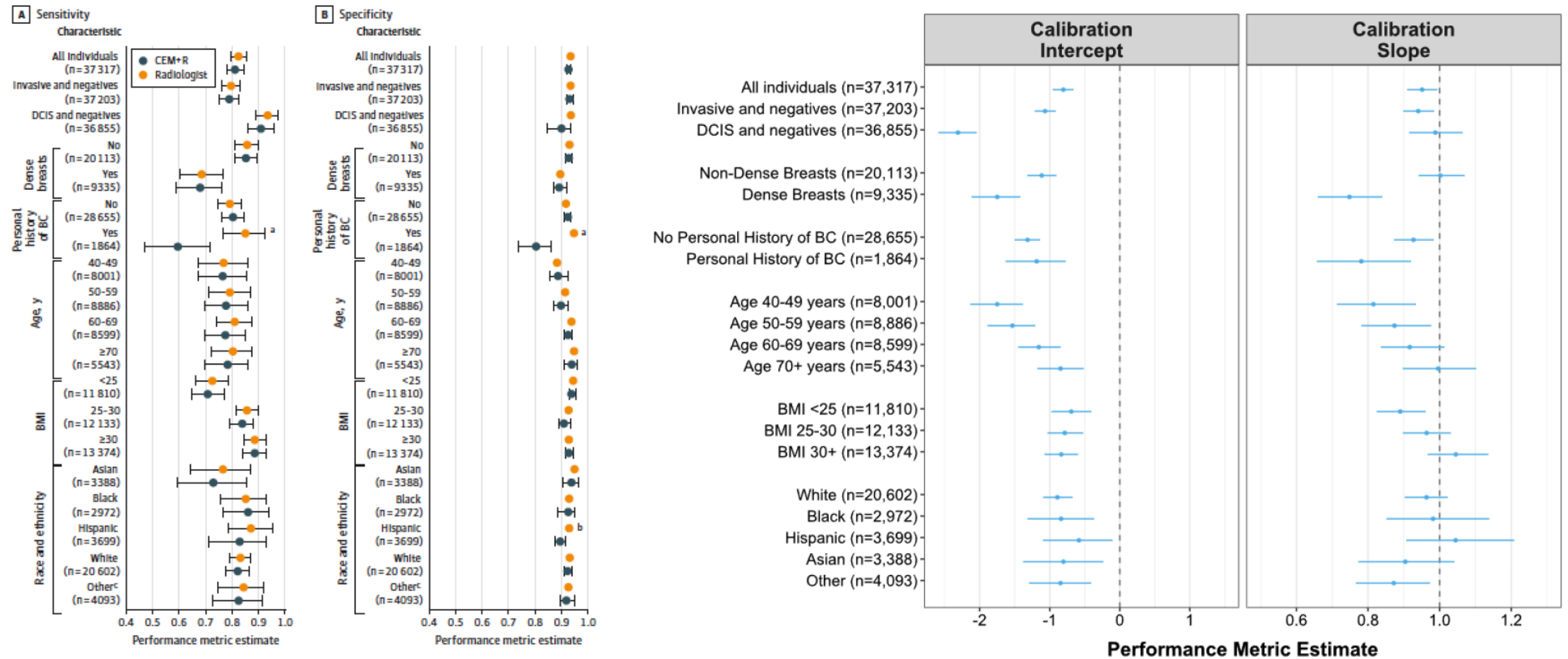


Example: External validation

Characteristic	Full cohort, unweighted	Analyzed subset ^a Unweighted, No. (%)
Women		
Total, No.	41 343	26 817
Diagnosed with BC within 12 mo of mammogram	714 (1.7)	573 (2.1)
Screening examinations		
Total, No.	121 753	37 317
BI-RADS category		
0	8493 (6.9)	3809 (10.2)
1	46 093 (37.9)	14 063 (37.7)
2	66 632 (54.7)	19 315 (51.8)
3, 4, or 5	535 (0.4)	130 (0.3)
BC diagnosed within 12 mo	723 (0.6)	576 (1.5)
Invasive BC	576 (79.7)	462 (80.2)
DCIS	147 (20.3)	114 (19.8)
No BC diagnosed within 12 mo	121 030 (99.4)	36 741 (98.5)
Age, mean (SD), y	58.9 (11.6)	58.4 (11.5)
BMI, mean (SD)	28.4 (6.9)	29.0 (7.0)
Personal history of breast cancer ^b	3751 (6.1)	1864 (6.1)
Race and ethnicity ^c		
Asian	10 913 (10.3)	3338 (9.7)
Black	9558 (9.0)	2972 (8.6)
Hispanic	11 449 (10.8)	3699 (10.6)
White	61 703 (58.3)	20 602 (59.3)
Other ^d	12 254 (11.6)	4093 (11.8)



Example: External validation



Notes on external validation

- Select an independent dataset
- Make sure your dataset is representative of the target population
- Utilize performance metrics that are appropriate for the problem
 - Set an expectation for minimum performance to adopt
- Process the data consistently, as was performed during model training
- Utilize appropriate statistical tests to compare the performance of the model on the external dataset with its performance on the training/validation datasets
- Check if algorithm performs consistently across subgroups
- Use results as a data point along with other clinical factors when making a “go”/“no go” decision on adopting algorithm

Resources

- Contact me
 - William Hsu, PhD whsu@mednet.ucla.edu
 - <http://www.hsu-lab.com>
- Electronic course notes and supplementary materials
 - <https://uclawillhsu.github.io/spie-tutorial/>



Brief read

- Liu Y, Chen PH, Krause J, Peng L. How to read articles that use machine learning: users' guides to the medical literature. Jama. 2019 Nov 12;322(18):1806-16.

More in-depth

- Steyerberg, EW. Evaluation of Performance. In: Clinical Prediction Models. Statistics for Biology and Health. Springer, Cham; 2019.

Textbook reference

- Japkowicz N, Shah M. Evaluating learning algorithms: a classification perspective. Cambridge University Press; 2011.

Questions?